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Surds and Indices

IMPORTANT FACTS AND FORMULAE

I. Laws of Indices:

$$(i) a^m \times a^n = a^{m+n}$$

$$(ii) \frac{a^m}{a^n} = a^{m-n}$$

$$(iii) (a^m)^n = a^{mn}$$

$$(iv) (ab)^n = a^n b^n$$

$$(v) \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$(vi) a^0 = 1$$

II. Surds: Let a be a rational number and n be a positive integer such that $a^{\frac{1}{n}} = \sqrt[n]{a}$ is irrational. Then, $\sqrt[n]{a}$ is called a surd of order n .

III. Laws of Surds:

$$(i) \sqrt[n]{a} = a^{\frac{1}{n}}$$

$$(ii) \sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}$$

$$(iii) \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$(iv) (\sqrt[n]{a})^n = a$$

$$(v) \sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$$

$$(vi) (\sqrt[n]{a})^m = \sqrt[n]{a^m}$$

SOLVED EXAMPLES

Ex. 1. Simplify : (i) $(27)^{\frac{2}{3}}$ (ii) $(1024)^{-\frac{4}{5}}$ (iii) $\left(\frac{8}{125}\right)^{-\frac{4}{3}}$.

Sol. (i) $(27)^{\frac{2}{3}} = (3^3)^{\frac{2}{3}} = 3^{(3 \times \frac{2}{3})} = 3^2 = 9$.

$$(ii) (1024)^{-\frac{4}{5}} = (4^5)^{-\frac{4}{5}} = 4^{\left\{5 \times \left(-\frac{4}{5}\right)\right\}} = 4^{-4} = \frac{1}{4^4} = \frac{1}{256}.$$

$$(iii) \left(\frac{8}{125}\right)^{-\frac{4}{3}} = \left\{\left(\frac{2}{5}\right)^3\right\}^{-\frac{4}{3}} = \left(\frac{2}{5}\right)^{\left\{3 \times \left(-\frac{4}{3}\right)\right\}} = \left(\frac{2}{5}\right)^{-4} = \left(\frac{5}{2}\right)^4 = \frac{5^4}{2^4} = \frac{625}{16}.$$

Ex. 2. What will come in place of both the question marks in the following question?

(Bank Recruitment, 2010)

$$\frac{(\frac{1}{?})^4}{(\frac{1}{?})^4} = \frac{48}{\frac{3}{4}}$$

Sol. Let $\frac{1}{x^4} = \frac{48}{\frac{3}{4}}$. Then, $x^{\frac{1}{4}} \cdot x^{\frac{3}{4}} = 48 \Leftrightarrow x^{\left(\frac{1}{4} + \frac{3}{4}\right)} = 48 \Leftrightarrow x = 48$.

Ex. 3. Evaluate : (i) $(.00032)^{\frac{3}{5}}$ (ii) $(256)^{0.16} \times (16)^{0.18}$.

Sol. (i) $(0.00032)^{\frac{3}{5}} = \left(\frac{32}{100000}\right)^{\frac{3}{5}} = \left(\frac{2^5}{10^5}\right)^{\frac{3}{5}} = \left\{\left(\frac{2}{10}\right)^5\right\}^{\frac{3}{5}} = \left(\frac{1}{5}\right)^{(5 \times \frac{3}{5})} = \left(\frac{1}{5}\right)^3 = \frac{1}{125}.$

$$\begin{aligned}
 (ii) \quad (256)^{0.16} \times (16)^{0.18} &= \{(16)^2\}^{0.16} \times (16)^{0.18} = (16)^{(2 \times 0.16)} \times (16)^{0.18} \\
 &= (16)^{0.32} \times (16)^{0.18} = (16)^{(0.32+0.18)} = (16)^{0.5} = (16)^{\frac{1}{2}} = 4.
 \end{aligned}$$

Ex. 4. Solve : $9^{8.6} \times 8^{3.9} \times 72^{4.4} \times 9^{3.9} \times 8^{8.6} = 72^x$.

(L.I.C., 2005)

Sol. Let $9^{8.6} \times 8^{3.9} \times 72^{4.4} \times 9^{3.9} \times 8^{8.6} = 72^x$.

$$\text{Then, } 9^{(8.6+3.9)} \times 8^{(3.9+8.6)} \times 72^{4.4} = 72^x$$

$$\Leftrightarrow 9^{12.5} \times 8^{12.5} \times 72^{4.4} = 72^x \Leftrightarrow (9 \times 8)^{12.5} \times 72^{4.4} = 72^x$$

$$\Leftrightarrow 72^{12.5} \times 72^{4.4} = 72^x \Leftrightarrow 72^{(12.5+4.4)} = 72^x \Leftrightarrow 72^{16.9} = 72^x \Leftrightarrow x = 16.9.$$

Ex. 5. Solve : $(0.064) \times (0.4)^7 = (0.4)^x \times (0.0256)^2$.

(Bank P.O., 2010)

Sol. Let $(0.064) \times (0.4)^7 = (0.4)^x \times (0.0256)^2$.

$$\text{Then, } (0.4)^3 \times (0.4)^7 = (0.4)^x \times [(0.4)^4]^2$$

$$\Leftrightarrow (0.4)^{(3+7)} = (0.4)^x \times (0.4)^8 \Leftrightarrow (0.4)^{10} = (0.4)^{x+8} \Leftrightarrow x+8=10 \Leftrightarrow x=2.$$

Ex. 6. What is the quotient when $(x^{-1} - 1)$ is divided by $(x - 1)$?

Sol.
$$\frac{x^{-1} - 1}{x - 1} = \frac{\frac{1}{x} - 1}{x - 1} = \frac{(1-x)}{x} \times \frac{1}{(x-1)} = -\frac{1}{x}.$$

Hence, the required quotient is $-\frac{1}{x}$.

Ex. 7. If $2^{x-1} + 2^{x+1} = 1280$, then find the value of x .

Sol. $2^{x-1} + 2^{x+1} = 1280 \Leftrightarrow 2^{x-1} (1 + 2^2) = 1280$

$$\Leftrightarrow 2^{x-1} = \frac{1280}{5} = 256 = 2^8 \Leftrightarrow x-1=8 \Leftrightarrow x=9.$$

Hence, $x = 9$.

Ex. 8. Find the value of $\left[5 \left(8^{\frac{1}{3}} + 27^{\frac{1}{3}}\right)^3\right]^{\frac{1}{4}}$.

Sol.
$$\begin{aligned} \left[5 \left(8^{\frac{1}{3}} + 27^{\frac{1}{3}}\right)^3\right]^{\frac{1}{4}} &= \left[5 \left\{(2^3)^{\frac{1}{3}} + (3^3)^{\frac{1}{3}}\right\}^3\right]^{\frac{1}{4}} = \left[5 \left\{2^{\left(3 \times \frac{1}{3}\right)} + 3^{\left(3 \times \frac{1}{3}\right)}\right\}^3\right]^{\frac{1}{4}} \\ &= \{5(2+3)^3\}^{\frac{1}{4}} = (5 \times 5^3)^{\frac{1}{4}} = (5^4)^{\frac{1}{4}} = 5^{\left(4 \times \frac{1}{4}\right)} = 5^1 = 5. \end{aligned}$$

Ex. 9. Find the value of $\left\{(16)^{\frac{3}{2}} + (16)^{-\frac{3}{2}}\right\}$.

Sol.
$$\begin{aligned} \left[(16)^{\frac{3}{2}} + (16)^{-\frac{3}{2}}\right] &= \left[(4^2)^{\frac{3}{2}} + (4^2)^{-\frac{3}{2}}\right] = 4^{\left(2 \times \frac{3}{2}\right)} + 4^{\left\{2 \times \left(-\frac{3}{2}\right)\right\}} \\ &= 4^3 + 4^{-3} = 4^3 + \frac{1}{4^3} = \left(64 + \frac{1}{64}\right) = \frac{4097}{64}. \end{aligned}$$

Ex. 10. If $\left(\frac{1}{5}\right)^{3y} = 0.008$, then find the value of $(0.25)^y$.

Sol.
$$\begin{aligned} \left(\frac{1}{5}\right)^{3y} = 0.008 &= \frac{8}{1000} = \frac{1}{125} = \left(\frac{1}{5}\right)^3 \Leftrightarrow 3y = 3 \Leftrightarrow y = 1. \\ \therefore (0.25)^y &= (0.25)^1 = 0.25. \end{aligned}$$

Ex. 11. Simplify : $\frac{(6.25)^{\frac{1}{2}} \times (0.0144)^{\frac{1}{2}} + 1}{(0.027)^{\frac{1}{3}} \times (81)^{\frac{1}{4}}}$.

(S.S.C., 2005)

Sol. Given expression = $\frac{\{(2.5)^2\}^{\frac{1}{2}} \times \{(0.12)^2\}^{\frac{1}{2}} + 1}{\{(0.3)^3\}^{\frac{1}{3}} \times (3^4)^{\frac{1}{4}}} = \frac{(2.5)^{\left(2 \times \frac{1}{2}\right)} \times (0.12)^{\left(2 \times \frac{1}{2}\right)} + 1}{(0.3)^{\left(3 \times \frac{1}{3}\right)} \times 3^{\left(4 \times \frac{1}{4}\right)}}$
 $= \frac{2.5 \times 0.12 + 1}{0.3 \times 3} = \frac{0.3 + 1}{0.9} = \frac{1.3}{0.9} = \frac{13}{9} = 1.444... = 1.\bar{4}$.

Ex. 12. Find the value of $\frac{(243)^{\frac{n}{5}} \cdot 3^{2n+1}}{9^n \times 3^{n-1}}$.

Sol. $\frac{(243)^{\frac{n}{5}} \cdot 3^{2n+1}}{9^n \times 3^{n-1}} = \frac{(3^5)^{\frac{n}{5}} \times 3^{2n+1}}{(3^2)^n \times 3^{n-1}} = \frac{3^{\left(5 \times \frac{n}{5}\right)} \times 3^{2n+1}}{3^{2n} \times 3^{n-1}} = \frac{3^n \times 3^{2n+1}}{3^{2n} \times 3^{n-1}}$
 $= \frac{3^{n+(2n+1)}}{3^{2n+n-1}} = \frac{3^{(3n+1)}}{3^{(3n-1)}} = 3^{(3n+1)-(3n-1)} = 3^2 = 9$.

Ex. 13. Find the value of $\left(2^{\frac{1}{4}} - 1\right) \left(2^{\frac{3}{4}} + 2^{\frac{1}{2}} + 2^{\frac{1}{4}} + 1\right)$.

(N.I.F.T., 2003)

Sol. Putting $2^{\frac{1}{4}} = x$, we get :

$$\begin{aligned} \left(2^{\frac{1}{4}} - 1\right) \left(2^{\frac{3}{4}} + 2^{\frac{1}{2}} + 2^{\frac{1}{4}} + 1\right) &= (x-1)(x^3 + x^2 + x + 1) \\ &= (x-1)[x^2(x+1) + (x+1)] \\ &= (x-1)(x+1)(x^2+1) = (x^2-1)(x^2+1) \\ &= (x^4-1) = \left[\left(2^{\frac{1}{4}}\right)^4 - 1\right] = \left[2^{\left(\frac{1}{4} \times 4\right)} - 1\right] = (2-1) = 1. \end{aligned}$$

Ex. 14. Find the value of $\frac{6^{\frac{2}{3}} \times \sqrt[3]{6^7}}{\sqrt[3]{6^6}}$.

Sol. $\frac{6^{\frac{2}{3}} \times \sqrt[3]{6^7}}{\sqrt[3]{6^6}} = \frac{6^{\frac{2}{3}} \times (6^7)^{\frac{1}{3}}}{(6^6)^{\frac{1}{3}}} = \frac{6^{\frac{2}{3}} \times 6^{\left(7 \times \frac{1}{3}\right)}}{6^{\left(6 \times \frac{1}{3}\right)}} = \frac{6^{\frac{2}{3}} \times 6^{\left(\frac{7}{3}\right)}}{6^2}$
 $= 6^{\frac{2}{3}} \times 6^{\left(\frac{7}{3}-2\right)} = 6^{\frac{2}{3}} \times 6^{\frac{1}{3}} = 6^{\left(\frac{2}{3}+\frac{1}{3}\right)} = 6^1 = 6$.

Ex. 15. If $\left(\frac{p}{q}\right)^{rx-s} = \left(\frac{q}{p}\right)^{px-q}$, then find the value of x .

Sol. $\left(\frac{p}{q}\right)^{rx-s} = \left(\frac{q}{p}\right)^{px-q} \Leftrightarrow \left(\frac{p}{q}\right)^{rx-s} = \left(\frac{p}{q}\right)^{-(px-q)}$
 $\Leftrightarrow rx-s = -(px-q) \Leftrightarrow rx-s = -px+q$
 $\Leftrightarrow rx+px = q+s \Leftrightarrow x(p+r) = q+s$
 $\Leftrightarrow x = \frac{q+s}{p+r}$.

Ex. 16. If $x = y^a$, $y = z^b$ and $z = x^c$, then find the value of abc .

Sol. $z^1 = x^c = (y^a)^c \quad [\because x = y^a]$
 $= y^{(ac)} = (z^b)^{ac} \quad [\because y = z^b]$
 $= z^{b(ac)} = z^{abc}$
 $\therefore abc = 1.$

Ex. 17. Simplify : $\left(\frac{x^a}{x^b}\right)^{(a^2+b^2+ab)} \times \left(\frac{x^b}{x^c}\right)^{(b^2+c^2+bc)} \times \left(\frac{x^c}{x^a}\right)^{(c^2+a^2+ca)}.$

Sol. Given Expression = $\{x^{(a-b)}\}^{(a^2+b^2+ab)} \cdot \{x^{(b-c)}\}^{(b^2+c^2+bc)} \cdot \{x^{(c-a)}\}^{(c^2+a^2+ca)}$
 $= x^{(a-b)(a^2+b^2+ab)} \cdot x^{(b-c)(b^2+c^2+bc)} \cdot x^{(c-a)(c^2+a^2+ca)}$
 $= x^{(a^3-b^3)} \cdot x^{(b^3-c^3)} \cdot x^{(c^3-a^3)} = x^{(a^3-b^3+b^3-c^3+c^3-a^3)} = x^0 = 1.$

Ex. 18. If $8^x \cdot 2^y = 512$ and $3^{3x+2y} = 9^6$, then what is the value of x and y ? (M.A.T., 2004)

Sol. $8^x \cdot 2^y = 512 \Leftrightarrow (2^3)^x \cdot 2^y = 2^9 \Leftrightarrow 2^{3x+y} = 2^9 \Leftrightarrow 3x+y=9$...(i)
 And, $3^{3x+2y} = 9^6 \Leftrightarrow 3^{3x+2y} = (3^2)^6 = 3^{12} \Leftrightarrow 3x+2y=12$...(ii)
 Subtracting (i) from (ii), we get: $y=3$.
 Putting $y=3$ in (i), we get: $3x=6$ or $x=2$.
 Hence, $x=2$ and $y=3$.

Ex. 19. Find the largest from among $\sqrt[4]{6}$, $\sqrt{2}$ and $\sqrt[3]{4}$.

Sol. Given surds are of order 4, 2 and 3 respectively. Their L.C.M. is 12.
 Changing each to a surd of order 12, we get :

$$\sqrt[4]{6} = 6^{\frac{1}{4}} = 6^{\left(\frac{1}{4} \times \frac{3}{3}\right)} = \left(6^{\frac{3}{12}}\right) = (6^3)^{\frac{1}{12}} = (216)^{\frac{1}{12}}.$$

$$\sqrt{2} = 2^{\frac{1}{2}} = 2^{\left(\frac{1}{2} \times \frac{6}{6}\right)} = \left(2^{\frac{6}{12}}\right) = (2^6)^{\frac{1}{12}} = (64)^{\frac{1}{12}}.$$

$$\sqrt[3]{4} = 4^{\frac{1}{3}} = 4^{\left(\frac{1}{3} \times \frac{4}{4}\right)} = \left(4^{\frac{4}{12}}\right) = (4^4)^{\frac{1}{12}} = (256)^{\frac{1}{12}}.$$

$$\text{Clearly, } (256)^{\frac{1}{12}} > (216)^{\frac{1}{12}} > (64)^{\frac{1}{12}}.$$

$$\therefore \text{Largest one is } (256)^{\frac{1}{12}} \text{ i.e., } \sqrt[3]{4}.$$

Ex. 20. Find the square root of $(3+\sqrt{5})$. (L.I.C.A.O., 2007)

Sol. $\sqrt{3+\sqrt{5}} = \sqrt{3+2\sqrt{\frac{5}{4}}} = \sqrt{\frac{5}{2} + \frac{1}{2} + 2\sqrt{\frac{5}{2} \times \frac{1}{2}}} = \sqrt{\left(\sqrt{\frac{5}{2}}\right)^2 + \left(\sqrt{\frac{1}{2}}\right)^2 + 2\sqrt{\frac{5}{2}}\sqrt{\frac{1}{2}}}$
 $= \sqrt{\left(\sqrt{\frac{5}{2}} + \sqrt{\frac{1}{2}}\right)^2} = \left(\sqrt{\frac{5}{2}} + \sqrt{\frac{1}{2}}\right).$

Ex. 21. If $x = 3+2\sqrt{2}$, find the value of $\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right).$

Sol. $\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 = x + \frac{1}{x} - 2 = (3+2\sqrt{2}) + \frac{1}{(3+2\sqrt{2})} - 2$
 $= (3+2\sqrt{2}) + \frac{1}{(3+2\sqrt{2})} \times \frac{(3-2\sqrt{2})}{(3-2\sqrt{2})} - 2 = (3+2\sqrt{2}) + (3-2\sqrt{2}) - 2 = 4.$
 $\therefore \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right) = 2.$

Ex. 22. If $2^x = 3^y = 6^{-z}$, find the value of $\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$.

Sol. Let $2^x = 3^y = 6^{-z} = k$. Then, $2 = k^{\frac{1}{x}}$, $3 = k^{\frac{1}{y}}$ and $6 = k^{-\frac{1}{z}}$.

$$\text{Now, } 2 \times 3 = 6 \Leftrightarrow k^{\frac{1}{x}} \times k^{\frac{1}{y}} = k^{-\frac{1}{z}} \Leftrightarrow k^{\left(\frac{1}{x} + \frac{1}{y}\right)} = k^{-\frac{1}{z}}$$

$$\Leftrightarrow \frac{1}{x} + \frac{1}{y} = -\frac{1}{z} \Leftrightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0.$$

Ex. 23. Given $t = 2 + \sqrt[3]{4} + \sqrt[3]{2}$, determine the value of $t^3 - 6t^2 + 6t - 2$.

(M.A.T., 2005)

Sol. $t = 2 + \sqrt[3]{4} + \sqrt[3]{2} \Rightarrow t - 2 = \sqrt[3]{4} + \sqrt[3]{2} \Rightarrow (t - 2)^3 = (\sqrt[3]{4} + \sqrt[3]{2})^3$

$$\Rightarrow t^3 - 8 - 6t(t - 2) = 4 + 2 + 3 \sqrt[3]{8}(\sqrt[3]{4} + \sqrt[3]{2})$$

$$\Rightarrow t^3 - 8 - 6t^2 + 12t = 6 + 3 \times 2(t - 2)$$

$$\Rightarrow t^3 - 6t^2 + 12t - 8 = 6 + 6t - 12 \Rightarrow t^3 - 6t^2 + 6t - 2 = 0.$$

EXERCISE

(OBJECTIVE TYPE QUESTIONS)

Directions: Mark (✓) against the correct answer:

1. $\sqrt[3]{5}$ is a surd of the order (R.R.B., 2008)

- (a) $\frac{1}{3}$ (b) 1
(c) 2 (d) 3

2. $5^0 \times 8 = ?$ (R.R.B., 2006)

- (a) 0 (b) 8
(c) 40 (d) 200

3. Which of the following are equal in value?

- I. 4^1 II. 1^4
III. 4^0 IV. 0^4
(a) I and II (b) II and III
(c) III and IV (d) I and IV

4. If $289 = 17^{\frac{1}{5-x}}$, then $x = ?$ (Bank P.O., 2009)

- (a) $\frac{2}{5}$ (b) 8
(c) 16 (d) 32
(e) None of these

5. $(81)^4 \div (9)^5 = ?$ (Agriculture Officers', 2009)

- (a) 9 (b) 81
(c) 729 (d) 6561
(e) None of these

6. The value of $\left(\frac{9^2 \times 18^4}{3^{16}}\right)$ is (R.R.B., 2006)

- (a) $\frac{3}{2}$ (b) $\frac{4}{9}$
(c) $\frac{16}{81}$ (d) $\frac{32}{243}$

7. $[4^3 \times 5^4] \div 4^5 = ?$

(Bank Recruitment, 2008)

- (a) 29.0825 (b) 30.0925
(c) 35.6015 (d) 39.0625
(e) None of these

8. $9^3 \times 6^2 \div 3^3 = ?$

(L.I.C.A.D.O., 2007)

- (a) 948 (b) 972
(c) 984 (d) 1012
(e) None of these

9. $(19)^{12} \times (19)^8 \div (19)^4 = (19)^?$ (Bank Recruitment, 2008)

- (a) 6 (b) 8
(c) 12 (d) 24
(e) None of these

10. $(64)^4 \div (8)^5 = ?$

(Agriculture Officer's, 2008)

- (a) $(8)^8$ (b) $(8)^2$
(c) $(8)^{12}$ (d) $(8)^4$
(e) None of these

11. $(1000)^{12} \div (10)^{30} = ?$

(Bank P.O., 2008)

- (a) $(1000)^2$ (b) 10
(c) 100 (d) $(100)^2$
(e) None of these

12. $(3)^8 \times (3)^4 = ?$

(Bank P.O., 2009)

- (a) $(27)^3$ (b) $(27)^5$
(c) $(729)^2$ (d) $(729)^3$
(e) None of these

13. $\frac{343 \times 49}{216 \times 16 \times 81} = ?$

(Bank P.O., 2010)

- (a) $\frac{7^5}{6^7}$ (b) $\frac{7^5}{6^8}$
(c) $\frac{7^6}{6^7}$ (d) $\frac{7^4}{6^8}$
(e) None of these

14. $\frac{16 \times 32}{9 \times 27 \times 81} = ?$ (Bank P.O., 2009)
- (a) $\left(\frac{2}{3}\right)^9$ (b) $\left(\frac{2}{3}\right)^{11}$
 (c) $\left(\frac{2}{3}\right)^{12}$ (d) $\left(\frac{2}{3}\right)^{13}$
 (e) None of these
15. $9^3 \times (81)^2 \div (27)^3 = (3)^?$ (Bank P.O., 2010)
- (a) 3 (b) 4
 (c) 5 (d) 6
 (e) None of these
16. $(6)^4 \div (36)^3 \times 216 = 6^{(? - 5)}$ (Bank Recruitment, 2010)
- (a) 1 (b) 4
 (c) 6 (d) 7
 (e) None of these
17. $(0.2)^2, \frac{1}{100}, (0.01)^2, (0.008)^3$. Of these, which one is the greatest? (P.C.S., 2004)
- (a) $(0.008)^3$ (b) $(0.01)^2$
 (c) $(0.2)^2$ (d) $\frac{1}{100}$
18. Which of the following expressions has the greatest value?
- (a) $[(2^{-1})^0]^2$ (b) $\left[(4^0)^{-\frac{1}{2}}\right]^2$
 (c) $[(2^{-2})^{-1}]^2$ (d) $[(2^{-1})^2]^2$
19. $(10)^{24} \times (10)^{-21} = ?$ (Bank Recruitment, 2008)
- (a) 3 (b) 10
 (c) 100 (d) 1000
 (e) None of these
20. The value of $(256)^{\frac{5}{4}}$ is
- (a) 512 (b) 984
 (c) 1024 (d) 1032
21. The value of $(\sqrt{8})^{\frac{1}{3}}$ is
- (a) 2 (b) 4
 (c) $\sqrt{2}$ (d) 8
22. The value of $\left(\frac{32}{243}\right)^{-\frac{4}{5}}$ is
- (a) $\frac{4}{9}$ (b) $\frac{9}{4}$
 (c) $\frac{16}{81}$ (d) $\frac{81}{16}$
23. The value of $\left(-\frac{1}{216}\right)^{-\frac{2}{3}}$ is
- (a) 36 (b) -36
 (c) $\frac{1}{36}$ (d) $-\frac{1}{36}$
24. The value of $27^{-\frac{2}{3}}$ lies between (C.D.S., 2002)
- (a) 0 and 1 (b) 1 and 2
 (c) 2 and 3 (d) 3 and 4
25. The value of $\sqrt[3]{2^4 \sqrt{2^{-5} \sqrt{2^6}}}$ is (S.S.C., 2005)
- (a) 1 (b) 2
 (c) $2^{\frac{5}{3}}$ (d) 2^5
26. $\sqrt{2\sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}}$ = ? (R.R.B., 2007)
- (a) $2^{\frac{29}{31}}$ (b) $2^{\frac{31}{32}}$
 (c) $2^{\frac{9}{2}}$ (d) $2^{\frac{11}{2}}$
27. The value of $(0.03125)^{-\frac{2}{5}}$ is (R.R.B., 2006)
- (a) 4 (b) 9
 (c) 12 (d) 31.25
28. $\left(\frac{1}{2}\right)^{-\frac{1}{2}}$ is equal to (Section Officer's, 2005)
- (a) $\frac{1}{\sqrt{2}}$ (b) $2\sqrt{2}$
 (c) $-\sqrt{2}$ (d) $\sqrt{2}$
29. Simplified form of $\left[\left(\sqrt[5]{x^{-\frac{3}{5}}}\right)^{-\frac{5}{3}}\right]^5$ is (S.S.C., 2010)
- (a) $\frac{1}{x}$ (b) x
 (c) x^{-5} (d) x^5
30. What will come in place of both the question marks in the following question? (Bank Recruitment, 2010)
- $$\frac{(\frac{2}{3})^3}{42} = \frac{5}{(\frac{1}{?})^3}$$
- (a) 10 (b) $10\sqrt{2}$
 (c) $\sqrt{20}$ (d) 20
 (e) 210
31. The value of $5^{\frac{1}{4}} \times (125)^{0.25}$ is :
- (a) $\sqrt{5}$ (b) 5
 (c) $5\sqrt{5}$ (d) 25

32. The value of $\frac{1}{(216)^{-\frac{2}{3}}} + \frac{1}{(256)^{-\frac{3}{4}}} + \frac{1}{(32)^{-\frac{1}{5}}}$ is

(M.B.A., 2003)

- (a) 102 (b) 105
(c) 107 (d) 109

33. $(2.4 \times 10^3) \div (8 \times 10^{-2}) = ?$

- (a) 3×10^{-5} (b) 3×10^4
(c) 3×10^5 (d) 30

34. $\left(\frac{1}{216}\right)^{-\frac{2}{3}} \div \left(\frac{1}{27}\right)^{-\frac{4}{3}} = ?$

- (a) $\frac{3}{4}$ (b) $\frac{2}{3}$
(c) $\frac{4}{9}$ (d) $\frac{1}{8}$

35. $(48)^{-\frac{2}{7}} \times (16)^{-\frac{5}{7}} \times (3)^{-\frac{5}{7}} = ?$

(P.C.S., 2008)

- (a) $\frac{1}{3}$ (b) $\frac{1}{48}$
(c) 1 (d) 48

36. If $10^x = \frac{1}{2}$, then $10^{-8x} = ?$

(P.C.S., 2008)

- (a) $\frac{1}{256}$ (b) 16
(c) 80 (d) 256

37. If $\left(\frac{3}{5}\right)^3 \left(\frac{3}{5}\right)^{-6} = \left(\frac{3}{5}\right)^{2x-1}$, then x is equal to

(S.S.C., 2010)

- (a) -2 (b) -1
(c) 1 (d) 2

38. $49 \times 49 \times 49 \times 49 = 7^?$

- (a) 4 (b) 7
(c) 8 (d) 16

39. The value of $(8^{-25} - 8^{-26})$ is

- (a) 7×8^{-25} (b) 7×8^{-26}
(c) 8×8^{-26} (d) None of these

40. $(64)^{-\frac{1}{2}} - (-32)^{-\frac{4}{5}} = ?$

- (a) $\frac{1}{8}$ (b) $\frac{3}{8}$
(c) $\frac{1}{16}$ (d) $\frac{3}{16}$
(e) None of these

41. If $\left(\frac{a}{b}\right)^{x-1} = \left(\frac{b}{a}\right)^{x-3}$, then the value of x is

(P.C.S., 2009)

- (a) $\frac{1}{2}$ (b) 1
(c) 2 (d) $\frac{7}{2}$

42. If $2^{2n-1} = \frac{1}{8^{n-3}}$, then the value of n is

- (a) -2 (b) 0
(c) 2 (d) 3

43. If $5^a = 3125$, then the value of $5^{(a-3)}$ is

- (a) 25 (b) 125
(c) 625 (d) 1625

44. If $5\sqrt{5} \times 5^3 \div 5^{-\frac{3}{2}} = 5^{a+2}$, then the value of a is

(M.B.A., 2006)

- (a) 4 (b) 5
(c) 6 (d) 8

45. If $\sqrt{2^n} = 64$, then the value of n is

- (a) 2 (b) 4
(c) 6 (d) 12

46. If $(\sqrt{3})^5 \times 9^2 = 3^n \times 3\sqrt{3}$, then the value of n is

- (a) 2 (b) 3
(c) 4 (d) 5

47. If $\frac{9^n \times 3^5 \times (27)^3}{3 \times (81)^4} = 27$, then the value of n is

- (a) 0 (b) 2
(c) 3 (d) 4

48. If $\left(\frac{9}{4}\right)^x \cdot \left(\frac{8}{27}\right)^{x-1} = \frac{2}{3}$, then the value of x is

- (a) 1 (b) 2
(c) 3 (d) 4

49. If $2^x = \sqrt[3]{32}$, then x is equal to

- (a) 5 (b) 3
(c) $\frac{3}{5}$ (d) $\frac{5}{3}$

50. If $2^x \times 8^{\frac{1}{5}} = 2^{\frac{1}{5}}$, then x is equal to

- (a) $\frac{1}{5}$ (b) $-\frac{1}{5}$
(c) $\frac{2}{5}$ (d) $-\frac{2}{5}$

51. If $5^{(x+3)} = 25^{(3x-4)}$, then the value of x is

- (a) $\frac{5}{11}$ (b) $\frac{11}{5}$
(c) $\frac{11}{3}$ (d) $\frac{13}{5}$

52. $\frac{2^{n+4} - 2(2^n)}{2(2^{n+3})}$ when simplified is (M.B.A., 2011)

- (a) $2^{n+1} - \frac{1}{8}$ (b) -2^{n+1}
(c) $1 - 2^n$ (d) $\frac{7}{8}$

53. Simplify $\left[\sqrt[3]{\sqrt[6]{a^9}}\right]^4 \left[\sqrt[6]{\sqrt[3]{a^9}}\right]^4$; the result is (M.B.A., 2011)
 (a) a^4 (b) a^8
 (c) a^{12} (d) a^{16}
54. $(256)^{0.16} \times (256)^{0.09} = ?$ (S.S.C., 2004)
 (a) 4 (b) 16
 (c) 64 (d) 256.25
55. $(0.04)^{-1.5} = ?$ (Bank P.O., 2003)
 (a) 25 (b) 125
 (c) 250 (d) 625
56. $(17)^{3.5} \times (17)^? = 17^8$ (Bank P.O., 2003)
 (a) 2.29 (b) 2.75
 (c) 4.25 (d) 4.5
57. $6^{1.2} \times 36^? \times 30^{2.4} \times 25^{1.3} = 30^5$ (Specialist Officers', 2006)
 (a) 0.1 (b) 0.7
 (c) 1.4 (d) 2.6
 (e) None of these
58. $2^{3.6} \times 4^{3.6} \times 4^{3.6} \times (32)^{2.3} = (32)^?$ (Specialist Officers', 2007)
 (a) 5.9 (b) 7.7
 (c) 9.5 (d) 13.1
 (e) None of these
59. $3^{3.5} \times 2^{12} \times 4^{2.5} \div 2^{2.5} \times 7^{3.5} = 21^?$ (Bank P.O., 2006)
 (a) 6.5 (b) 8
 (c) 10 (d) 12.5
 (e) None of these
60. $8^{0.4} \times 4^{1.6} \times 2^{1.6} = ?$ (Agriculture Officers', 2009)
 (a) 48 (b) 52
 (c) 64 (d) 76
 (e) None of these
61. $8^7 \times 2^6 \div 8^{2.4} = 8^?$ (Bank P.O., 2009)
 (a) 6.6 (b) 8.6
 (c) 9.6 (d) 10.6
 (e) None of these
62. $25^{2.7} \times 5^{4.2} \div 5^{5.4} = 25^?$ (Bank Recruitment, 2010)
 (a) 1.6 (b) 1.7
 (c) 3.2 (d) 3.6
 (e) None of these
63. $8^{2.4} \times 2^{3.7} - (16)^{1.3} = 2^?$ (Bank Recruitment, 2010)
 (a) 4.8 (b) 5.7
 (c) 5.8 (d) 7.1
 (e) None of these
64. $(0.04)^2 \div (0.008) \times (0.2)^6 = (0.2)^?$ (Bank Recruitment, 2010)
 (a) 5 (b) 6
 (c) 8 (d) 9
 (e) None of these
65. $(18)^{3.5} \div (27)^{3.5} \times 6^{3.5} = 2^?$ (Bank P.O., 2003)
 (a) 3.5 (b) 4.5
 (c) 6 (d) 7
 (e) None of these
66. $(25)^{7.5} \times (5)^{2.5} \div (125)^{1.5} = 5^?$ (Bank P.O., 2003)
 (a) 8.5 (b) 13
 (c) 16 (d) 17.5
 (e) None of these
67. The value of $\frac{(243)^{0.13} \times (243)^{0.07}}{(7)^{0.25} \times (49)^{0.075} \times (343)^{0.2}}$ is (C.B.I., 2003)
 (a) $\frac{3}{7}$ (b) $\frac{7}{3}$
 (c) $1\frac{3}{7}$ (d) $2\frac{2}{7}$
68. $(64x^3 + 27a^{-3})^{-\frac{2}{3}} = ?$ (R.R.B., 2006)
 (a) $\frac{9ax}{16}$ (b) $\frac{9}{16ax}$
 (c) $\frac{9}{16x^2a^2}$ (d) $\frac{3}{4}x^{-2}a^{-2}$
69. If $2^{n+4} - 2^{n+2} = 3$, then n is equal to
 (a) 0 (b) 2
 (c) -1 (d) -2
70. If $2^{n-1} + 2^{n+1} = 320$, then n is equal to
 (a) 6 (b) 8
 (c) 5 (d) 7
71. If $3^x - 3^{x-1} = 18$, then the value of x^x is
 (a) 3 (b) 8
 (c) 27 (d) 216
72. $\frac{2^{n+4} - 2 \times 2^n}{2 \times 2^{(n+3)}} + 2^{-3}$ is equal to
 (a) 2^{n+1} (b) $\left(\frac{9}{8} - 2^n\right)$
 (c) $\left(-2^{n+1} + \frac{1}{8}\right)$ (d) 1
73. The value of $\frac{2^{3x+4} + 8^{x+1}}{8^{x+1} - 2^{3x+2}}$ is
 (a) 3 (b) 4
 (c) 5 (d) 6
74. The value of $\frac{2^{n-1} - 2^n}{2^{n+4} + 2^{n+1}}$ is
 (a) $-\frac{1}{36}$ (b) $\frac{2}{3}$
 (c) $\frac{1}{13}$ (d) $\frac{5}{13}$

75. If $x = 5 + 2\sqrt{6}$, then $\sqrt{x} - \frac{1}{\sqrt{x}}$ is (A.A.O. Exam, 2009)
- (a) $2\sqrt{2}$ (b) $2\sqrt{3}$
(c) $\sqrt{3} + \sqrt{2}$ (d) $\sqrt{3} - \sqrt{2}$
76. $(4 + \sqrt{7})$, expressed as a perfect square, is equal to (Section Officers', 2005)
- (a) $(2 + \sqrt{7})^2$ (b) $\left(\frac{\sqrt{7}}{2} + \frac{1}{2}\right)^2$
(c) $\left\{\frac{1}{2}(\sqrt{7} + 1)^2\right\}$ (d) $(\sqrt{3} + \sqrt{4})^2$
77. $\sqrt{8 - 2\sqrt{15}}$ is equal to (C.P.O., 2007)
- (a) $3 - \sqrt{5}$ (b) $\sqrt{5} - \sqrt{3}$
(c) $5 - \sqrt{3}$ (d) $\sqrt{5} + \sqrt{3}$
78. $\sqrt{6 - 4\sqrt{3}} + \sqrt{16 - 8\sqrt{3}}$ is equal to (A.A.O. Exam, 2010)
- (a) $1 - \sqrt{3}$ (b) $\sqrt{3} - 1$
(c) $2(2 - \sqrt{3})$ (d) $2(2 + \sqrt{3})$
79. The value of $\frac{1}{\sqrt{12 - \sqrt{140}}} - \frac{1}{\sqrt{8 - \sqrt{60}}} - \frac{2}{\sqrt{10 + \sqrt{84}}}$ is (S.S.C., 2005)
- (a) 0 (b) 1
(c) 2 (d) 3
80. The value of the expression $\sqrt{4 + \sqrt{15}} + \sqrt{4 - \sqrt{15}} - \sqrt{12 - 4\sqrt{5}}$ is
- (a) an irrational number
(b) a negative integer
(c) a natural number
(d) a non-integer rational number
81. If $N = \frac{\sqrt{\sqrt{5} + 2} + \sqrt{\sqrt{5} - 2}}{\sqrt{\sqrt{5} + 1}} - \sqrt{3 - 2\sqrt{2}}$, then the value of N is (A.A.O. Exam., 2009)
- (a) $2\sqrt{2} - 1$ (b) 3
(c) 1 (d) 2
82. Given that $10^{0.48} = x$, $10^{0.70} = y$ and $x^z = y^2$, then the value of z is close to
- (a) 1.45 (b) 1.88
(c) 2.9 (d) 3.7
83. If m and n are whole numbers such that $m^n = 121$, then the value of $(m - 1)^{n+1}$ is (S.S.C., 2001)
- (a) 1 (b) 10
(c) 121 (d) 1000
84. Number of prime factors in $(216)^{\frac{3}{5}} \times (2500)^{\frac{2}{5}} \times (300)^{\frac{1}{5}}$ is
- (a) 6 (b) 7
(c) 8 (d) None of these
85. Number of prime factors in $\frac{6^{12} \times (35)^{28} \times (15)^{16}}{(14)^{12} \times (21)^{11}}$ is
- (a) 56 (b) 66
(c) 112 (d) None of these
86. $1 + (3 + 1)(3^2 + 1)(3^4 + 1)(3^8 + 1)(3^{16} + 1)(3^{32} + 1)$ is equal to (Section Officers', 2005)
- (a) $\frac{3^{64} - 1}{2}$ (b) $\frac{3^{64} + 1}{2}$
(c) $3^{64} - 1$ (d) $3^{64} + 1$
87. $\frac{1}{1 + a^{(n-m)}} + \frac{1}{1 + a^{(m-n)}} = ?$ (M.B.A., 2003; NMAT, 2006)
- (a) 0 (b) $\frac{1}{2}$
(c) 1 (d) $a^m + n$
88. If $a + b + c = 0$, then the value of $(x^a)^{a^2 - bc} \cdot (x^b)^{b^2 - ca} \cdot (x^c)^{c^2 - ab}$ is equal to
- (a) -2 (b) -1
(c) 0 (d) 1
89. $\frac{1}{1 + x^{(b-a)} + x^{(c-a)}} + \frac{1}{1 + x^{(a-b)} + x^{(c-b)}} + \frac{1}{1 + x^{(b-c)} + x^{(a-c)}} = ?$ (M.B.A., 2003)
- (a) 0 (b) 1
(c) $x^a - b - c$ (d) None of these
90. $\left(\frac{x^b}{x^c}\right)^{(b+c-a)} \cdot \left(\frac{x^c}{x^a}\right)^{(c+a-b)} \cdot \left(\frac{x^a}{x^b}\right)^{(a+b-c)} = ?$ (NMAT, 2005; L.I.C., 2003)
- (a) x^{abc} (b) 1
(c) $x^{ab+bc+ca}$ (d) $x^a + b + c$
91. $\left(\frac{x^a}{x^b}\right)^{(a+b)} \cdot \left(\frac{x^b}{x^c}\right)^{(b+c)} \cdot \left(\frac{x^c}{x^a}\right)^{(c+a)} = ?$ (M.B.A., 2006)
- (a) 0 (b) x^{abc}
(c) $x^a + b + c$ (d) 1
92. $\left(\frac{x^a}{x^b}\right)^{\frac{1}{ab}} \cdot \left(\frac{x^b}{x^c}\right)^{\frac{1}{bc}} \cdot \left(\frac{x^c}{x^a}\right)^{\frac{1}{ca}} = ?$
- (a) 1 (b) $\frac{1}{x^{abc}}$
(c) $x^{\frac{1}{(ab+bc+ca)}}$ (d) None of these

93. The expression $\frac{\left(x + \frac{1}{y}\right)^a \cdot \left(x - \frac{1}{y}\right)^b}{\left(y + \frac{1}{x}\right)^a \cdot \left(y - \frac{1}{x}\right)^b}$ reduces to
- (a) $\left(\frac{x}{y}\right)^{a-b}$ (b) $\left(\frac{y}{x}\right)^{a-b}$
 (c) $\left(\frac{x}{y}\right)^{a+b}$ (d) $\left(\frac{y}{x}\right)^{a+b}$
94. The value of $\left(x^{\frac{b+c}{c-a}}\right)^{\frac{1}{a-b}} \cdot \left(x^{\frac{c+a}{a-b}}\right)^{\frac{1}{b-c}} \cdot \left(x^{\frac{a+b}{b-c}}\right)^{\frac{1}{c-a}}$ is
- (a) 1 (b) a
 (c) b (d) c
95. If $x^p = y^q = z^r$ and $xyz = 1$, then the value of $p + q + r$ would be (M.B.A., 2008)
- (a) 0 (b) 1
 (c) 2 (d) a rational number
96. If $a^x = b^y = c^z$ and $b^2 = ac$, then y equals
- (a) $\frac{xz}{x+z}$ (b) $\frac{xz}{2(x-z)}$
 (c) $\frac{xz}{2(z-x)}$ (d) $\frac{2xz}{(x+z)}$
97. If $a^x = b$, $b^y = c$ and $c^z = a$, then the value of xyz is (M.B.A., 2005; R.R.B., 2008)
- (a) 0 (b) 1
 (c) $\frac{1}{abc}$ (d) abc
98. If $2^x = 4^y = 8^z$ and $\left(\frac{1}{2x} + \frac{1}{4y} + \frac{1}{6z}\right) = \frac{24}{7}$, then the value of z is
- (a) $\frac{7}{16}$ (b) $\frac{7}{32}$
 (c) $\frac{7}{48}$ (d) $\frac{7}{64}$
99. Suppose $4^a = 5$, $5^b = 6$, $6^c = 7$, $7^d = 8$, then the value of $abcd$ is (A.A.O. Exam, 2009)
- (a) 1 (b) $\frac{3}{2}$
 (c) 2 (d) $\frac{5}{2}$
100. If $abc = 1$, then $\left(\frac{1}{1+a+b^{-1}} + \frac{1}{1+b+c^{-1}} + \frac{1}{1+c+a^{-1}}\right) = ?$ (C.D.S. 2004)
- (a) 0 (b) 1
 (c) $\frac{1}{ab}$ (d) ab
101. If a, b, c are real numbers, then the value of $\sqrt{a^{-1}b} \cdot \sqrt{b^{-1}c} \cdot \sqrt{c^{-1}a}$ is
- (a) abc (b) $\sqrt{abc}$
 (c) $\frac{1}{abc}$ (d) 1
102. If $3^{(x-y)} = 27$ and $3^{(x+y)} = 243$, then x is equal to (R.R.B., 2003)
- (a) 0 (b) 2
 (c) 4 (d) 6
103. If $x^y = y^x$, then $\left(\frac{x}{y}\right)^{\frac{x}{y}}$ is equal to (M.C.A., 2005)
- (a) $\frac{y}{x^x}$ (b) $x^{\frac{x}{y}-1}$
 (c) 1 (d) $x^{\frac{x}{y}}$
104. If $4^{x+y} = 1$ and $4^{x-y} = 4$, then the values of x and y respectively are
- (a) $-\frac{1}{2}$ and $\frac{1}{2}$ (b) $-\frac{1}{2}$ and $-\frac{1}{2}$
 (c) $\frac{1}{2}$ and $-\frac{1}{2}$ (d) $\frac{1}{2}$ and $\frac{1}{2}$
105. If $2^{2x-1} + 4^x = 2^{x-\frac{1}{2}} + 2^{x+\frac{1}{2}}$, then x equals
- (a) $\frac{1}{2}$ (b) $\frac{2}{3}$
 (c) $\frac{3}{2}$ (d) 1
106. If $3^{2x-y} = 3^{x+y} = \sqrt{27}$, the value of y is (R.R.B., 2005)
- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$
 (c) $\frac{3}{2}$ (d) $\frac{3}{4}$
107. If $3^x = 5^y = 45^z$, then (C.D.S., 2002)
- (a) $\frac{2}{z} = \frac{1}{y} - \frac{1}{x}$ (b) $\frac{2}{y} = \frac{1}{x} - \frac{1}{z}$
 (c) $\frac{2}{x} = \frac{1}{z} - \frac{1}{y}$ (d) $x + y + z = 0$
108. Given $2^x = 8^{y+1}$ and $9^y = 3^{x-9}$, the value of $x + y$ is (M.B.A., 2010)
- (a) 18 (b) 21
 (c) 24 (d) 27
109. What are the values of x and y that satisfy the equation $2^{0.7x} \cdot 3^{-1.25y} = \frac{8\sqrt{6}}{27}$? (C.A.T., 2006)

- ## ANSWERS

- | | | | | | | | | | |
|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| 1. (d) | 2. (b) | 3. (b) | 4. (e) | 5. (c) | 6. (c) | 7. (d) | 8. (b) | 9. (e) | 10. (e) |
| 11. (a) | 12. (c) | 13. (a) | 14. (a) | 15. (c) | 16. (c) | 17. (a) | 18. (c) | 19. (d) | 20. (c) |
| 21. (c) | 22. (d) | 23. (a) | 24. (a) | 25. (b) | 26. (b) | 27. (a) | 28. (d) | 29. (b) | 30. (e) |
| 31. (b) | 32. (a) | 33. (b) | 34. (c) | 35. (b) | 36. (d) | 37. (b) | 38. (c) | 39. (b) | 40. (c) |
| 41. (c) | 42. (c) | 43. (a) | 44. (a) | 45. (d) | 46. (d) | 47. (c) | 48. (d) | 49. (d) | 50. (d) |
| 51. (b) | 52. (d) | 53. (a) | 54. (a) | 55. (b) | 56. (d) | 57. (b) | 58. (a) | 59. (b) | 60. (c) |
| 61. (a) | 62. (e) | 63. (b) | 64. (e) | 65. (d) | 66. (b) | 67. (a) | 68. (c) | 69. (d) | 70. (d) |
| 71. (c) | 72. (d) | 73. (d) | 74. (a) | 75. (a) | 76. (c) | 77. (b) | 78. (b) | 79. (a) | 80. (a) |
| 81. (c) | 82. (c) | 83. (d) | 84. (b) | 85. (b) | 86. (b) | 87. (c) | 88. (d) | 89. (b) | 90. (b) |
| 91. (d) | 92. (a) | 93. (c) | 94. (a) | 95. (a) | 96. (d) | 97. (b) | 98. (c) | 99. (b) | 100. (b) |
| 101. (d) | 102. (c) | 103. (b) | 104. (c) | 105. (a) | 106. (a) | 107. (c) | 108. (d) | 109. (d) | 110. (d) |
| 111. (b) | 112. (b) | 113. (b) | 114. (b) | 115. (b) | 116. (b) | 117. (d) | 118. (d) | 119. (d) | 120. (c) |
| 121. (a) | 122. (d) | | | | | | | | |

SOLUTIONS

1. $\sqrt[n]{a}$ is called a surd of order n .
2. $5^0 \times 8 = 1 \times 8 = 8$. [$\because 5^0 = 1$]
3. I. $4^1 = 4$ II. $1^4 = 1$
III. $4^0 = 1$ IV. $0^4 = 0$
4. $289 = 17^{\frac{1}{5}x} \Rightarrow 17^2 = 17^{\frac{1}{5}x} \Rightarrow \frac{1}{5}x = 2 \Rightarrow x = 2 \times 5 = 10$.
5. $(81)^4 \div (9)^5 = \frac{(9^2)^4}{(9^1)^5} = \frac{9^{(2 \times 4)}}{9^5} = \frac{9^8}{9^5} = 9^{(8-5)} = 9^3 = 729$.
6. $\left(\frac{9^2 \times 18^4}{3^{16}}\right) = \frac{9^2 \times (9 \times 2)^4}{3^{16}} = \frac{(3^2)^2 \times (3^2)^4 \times 2^4}{3^{16}}$

$$= \frac{3^4 \times 3^8 \times 2^4}{3^{16}} = \frac{3^{(4+8)} \times 2^4}{3^{16}} = \frac{3^{12} \times 2^4}{3^{16}}$$

$$= \frac{2^4}{3^{(16-12)}} = \frac{2^4}{3^4} = \frac{16}{81}$$
7. $\frac{4^3 \times 5^4}{4^5} = \frac{5^4}{4^{(5-3)}} = \frac{5^4}{4^2} = \frac{625}{16} = 39.0625$.
8. $\frac{9^3 \times 6^2}{3^3} = \frac{(3^2)^3 \times (3 \times 2)^2}{3^3} = \frac{3^{(2 \times 3)} \times 3^2 \times 2^2}{3^3} = \frac{3^{(6+2)} \times 2^2}{3^3}$

$$= \frac{3^{(8-3)} \times 2^2}{1} = 3^5 \times 2^2 = 243 \times 4 = 972$$
9. $\frac{(19)^{12} \times (19)^8}{(19)^4} = \frac{19^{(12+8)}}{(19)^4} = \frac{(19)^{20}}{(19)^4} = (19)^{(20-4)} = (19)^{16}$.
Hence, missing number = 16.
10. $(64)^4 \div (8)^5 = (8^2)^4 \div (8)^5 = (8)^{(2 \times 4)} \div 8^5 = \frac{8^8}{8^5} = 8^{(8-5)} = 8^3$.
11. $(1000)^{12} \div (10)^{30} = \frac{(10^3)^{12}}{(10)^{30}} = \frac{(10)^{(3 \times 12)}}{(10)^{30}} = \frac{(10)^{36}}{(10)^{30}} = (10)^{(36-30)}$

$$= 10^6 = (10^3)^2 = (1000)^2$$
12. $3^8 \times 3^4 = 3^{(8+4)} = 3^{12} = (3^6)^2 = (729)^2$.
13. $\frac{343 \times 49}{216 \times 16 \times 81} = \frac{7^3 \times 7^2}{6^3 \times 2^4 \times 3^4} = \frac{7^{(3+2)}}{6^3 \times (2 \times 3)^4}$

$$= \frac{7^5}{6^3 \times 6^4} = \frac{7^5}{6^{(3+4)}} = \frac{7^5}{6^7}$$
14. $\frac{16 \times 32}{9 \times 27 \times 81} = \frac{2^4 \times 2^5}{3^2 \times 3^3 \times 3^4} = \frac{2^{(4+5)}}{3^{(2+3+4)}} = \frac{2^9}{3^9} = \left(\frac{2}{3}\right)^9$.
15. Let $9^3 \times (81)^2 \div (27)^3 = 3^x$. Then,

$$3^x = \frac{(3^2)^3 \times (3^4)^2}{(3^3)^3} = \frac{3^{(2 \times 3)} \times 3^{(4 \times 2)}}{3^{(3 \times 3)}} = \frac{3^6 \times 3^8}{3^9} = \frac{3^{(6+8)}}{3^9}$$

$$\Rightarrow 3^x = \frac{3^{14}}{3^9} = 3^{(14-9)} = 3^5 \Rightarrow x = 5$$
16. Let $6^4 \div (36)^3 \times 216 = 6^{(x-5)}$.
Then, $6^{(x-5)} = 6^4 \div (6^2)^3 \times 6^3$

$$= 6^4 \div 6^{(2 \times 3)} \times 6^3 = 6^4 \div 6^6 \times 6^3 = 6^{(4-6+3)} = 6$$

$$\Rightarrow x - 5 = 1 \Rightarrow x = 6$$

17. $(0.2)^2 = 0.2 \times 0.2 = 0.04$;
 $\frac{1}{100} = 0.01$; $(0.01)^{\frac{1}{2}} = [(0.1)^2]^{\frac{1}{2}} = (0.1)^{(2 \times \frac{1}{2})} = 0.1$;
 $(0.008)^{\frac{1}{3}} = [(0.2)^3]^{\frac{1}{3}} = (0.2)^{(3 \times \frac{1}{3})} = 0.2$.
Clearly, $0.2 > 0.1 > 0.04 > 0.01$.
So, $(0.008)^{\frac{1}{3}}$ is the greatest.
18. $[(2^{-1})^0]^2 = \left[\left(\frac{1}{2}\right)^0\right]^2 = (1)^2 = 1$.
 $\left[(4^0)^{-\frac{1}{2}}\right]^2 = \left[(1)^{-\frac{1}{2}}\right]^2 = (1)^2 = 1$.
 $[(2^{-2})^{-1}]^2 = \left[\left(\frac{1}{2^2}\right)^{-1}\right]^2 = \left[\left(\frac{1}{4}\right)^{-1}\right]^2 = (4)^2 = 16$.
 $[(2^{-1})^2]^2 = \left[\left(\frac{1}{2}\right)^2\right]^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$.
19. $(10)^{24} \times (10)^{-21} = 10^{(24-21)} = 10^3 = 1000$.
20. $(256)^{\frac{5}{4}} = (4^4)^{\frac{5}{4}} = 4^{(4 \times \frac{5}{4})} = 4^5 = 1024$.
21. $(\sqrt{8})^{\frac{1}{3}} = \left(8^{\frac{1}{2}}\right)^{\frac{1}{3}} = 8^{\left(\frac{1}{2} \times \frac{1}{3}\right)}$

$$= 8^{\frac{1}{6}} = (2^3)^{\frac{1}{6}} = 2^{\left(3 \times \frac{1}{6}\right)} = 2^{\frac{1}{2}} = \sqrt{2}$$
22. $\left(\frac{32}{243}\right)^{-\frac{4}{5}} = \left\{\left(\frac{2}{3}\right)^5\right\}^{-\frac{4}{5}} = \left(\frac{2}{3}\right)^{5 \times \left(-\frac{4}{5}\right)}$

$$= \left(\frac{2}{3}\right)^{(-4)} = \left(\frac{3}{2}\right)^4 = \frac{3^4}{2^4} = \frac{81}{16}$$
23. $\left(-\frac{1}{216}\right)^{-\frac{2}{3}} = \left[\left(-\frac{1}{6}\right)^3\right]^{-\frac{2}{3}} = \left(-\frac{1}{6}\right)^{3 \times \left(-\frac{2}{3}\right)}$

$$= \left(-\frac{1}{6}\right)^{-2} = \frac{1}{\left(-\frac{1}{6}\right)^2} = \frac{1}{\left(\frac{1}{36}\right)} = 36$$
24. $(27)^{-\frac{2}{3}} \div (3^3)^{-\frac{2}{3}} = 3^{\left[3 \times \left(-\frac{2}{3}\right)\right]} = 3^{-2} = \frac{1}{3^2} = \frac{1}{9}$.
Clearly, $0 < \frac{1}{9} < 1$.
25. $\sqrt[3]{2^4 \sqrt{2^{-5}} \sqrt{2^6}} = \sqrt[3]{2^4 \sqrt{2^{-5}} (2^6)^{\frac{1}{2}}} = \sqrt[3]{2^4 \sqrt{2^{-5}} (2)^{(6 \times \frac{1}{2})}}$

$$= \sqrt[3]{2^4 \sqrt{2^{-5}} \cdot 2^3} = \sqrt[3]{2^4 \sqrt{2^{(-5+3)}}}$$

$$= \sqrt[3]{2^4 \sqrt{2^{(-2)}}} = \sqrt[3]{2^4 \cdot (2^{-2})^{\frac{1}{2}}} = \sqrt[3]{2^4 \cdot 2^{\left(-2 \times \frac{1}{2}\right)}}$$

$$\begin{aligned}
 &= \sqrt[3]{2^4 \cdot 2^{(-1)}} = \sqrt[3]{2^{(4-1)}} = \sqrt[3]{2^3} = (2^3)^{\frac{1}{3}} \\
 &= 2^{\left(3 \times \frac{1}{3}\right)} = 2.
 \end{aligned}$$

26. $\sqrt{2\sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}} = \sqrt{2\sqrt{2\sqrt{2\sqrt{2 \cdot 2^{\frac{1}{2}}}}}} = \sqrt{2\sqrt{2\sqrt{2\sqrt{2 \cdot 2^{\left(1+\frac{1}{2}\right)}}}}}$

$$\begin{aligned}
 &= \sqrt{2\sqrt{2\sqrt{2\sqrt{2 \cdot 2^{\frac{3}{2}}}}}} = \sqrt{2\sqrt{2\sqrt{2 \cdot \left(2^{\frac{3}{2}}\right)^{\frac{1}{2}}}}} \\
 &= \sqrt{2\sqrt{2\sqrt{2 \cdot 2^{\left(\frac{3}{2} \times \frac{1}{2}\right)}}}} = \sqrt{2\sqrt{2\sqrt{2 \cdot 2^{\frac{3}{4}}}}} \\
 &= \sqrt{2\sqrt{2\sqrt{2 \cdot 2^{\left(1+\frac{3}{4}\right)}}}} = \sqrt{2\sqrt{2\sqrt{2 \cdot 2^{\frac{7}{4}}}}} \\
 &= \sqrt{2\sqrt{2 \cdot \left(2^{\frac{7}{4}}\right)^{\frac{1}{2}}}} = \sqrt{2\sqrt{2 \cdot 2^{\left(\frac{7}{4} \times \frac{1}{2}\right)}}} \\
 &= \sqrt{2\sqrt{2 \cdot 2^{\frac{7}{8}}}} = \sqrt{2\sqrt{2 \cdot 2^{\left(1+\frac{7}{8}\right)}}} = \sqrt{2\sqrt{2 \cdot 2^{\frac{15}{8}}}} \\
 &= \sqrt{2 \cdot \left(2^{\frac{15}{8}}\right)^{\frac{1}{2}}} = \sqrt{2 \cdot 2^{\frac{15}{16}}} = \sqrt{2 \cdot 2^{\left(1+\frac{15}{16}\right)}} \\
 &= \sqrt{2 \cdot 2^{\frac{31}{16}}} = \left(2^{\frac{31}{16}}\right)^{\frac{1}{2}} = 2^{\left(\frac{31}{16} \times \frac{1}{2}\right)} = 2^{\frac{31}{32}}.
 \end{aligned}$$

27. $(0.03125)^{-\frac{2}{5}} = [(0.5)^5]^{-\frac{2}{5}} = 0.5^{\left[5 \times \left(-\frac{2}{5}\right)\right]} = (0.5)^{-2}$

$$= \frac{1}{(0.5)^2} = \frac{1}{0.25} = 4.$$

28. $\left(\frac{1}{2}\right)^{-\frac{1}{2}} = (2)^{\frac{1}{2}} = \sqrt{2}.$

29. $\left[\left(\sqrt[5]{x^{-\frac{3}{5}}}\right)^{-\frac{5}{3}}\right]^5 = \left[\left\{\left(x^{-\frac{3}{5}}\right)^{\frac{1}{5}}\right\}^{-\frac{5}{3}}\right]^5 = \left[\left\{x^{\left\{\left(-\frac{3}{5}\right) \times \frac{1}{5}\right\}}\right\}^{-\frac{5}{3}}\right]^5$

$$\begin{aligned}
 &= \left[\left(x^{-\frac{3}{25}}\right)^{-\frac{5}{3}}\right]^5 = \left[x^{\left\{\left(-\frac{3}{25}\right) \times \left(-\frac{5}{3}\right)\right\}}\right]^5 \\
 &= \left(x^{\frac{1}{5}}\right)^5 = x^{\left(\frac{1}{5} \times 5\right)} = x.
 \end{aligned}$$

30. Let $\frac{x^{\frac{2}{3}}}{42} = \frac{5}{x^{\frac{1}{3}}}.$

Then, $x^{\frac{2}{3}} \cdot x^{\frac{1}{3}} = 42 \times 5 = 210 \Rightarrow x^{\left(\frac{2}{3} + \frac{1}{3}\right)} = 210 \Rightarrow x = 210.$

31. $5^{\frac{1}{4}} \times (125)^{0.25} = 5^{0.25} \times (5^3)^{0.25} = 5^{0.25} \times 5^{(3 \times 0.25)}$

$$= 5^{0.25} \times 5^{0.75} = 5^{(0.25 + 0.75)} = 5^1 = 5.$$

32. $\frac{1}{(216)^{-\frac{2}{3}}} + \frac{1}{(256)^{-\frac{3}{4}}} + \frac{1}{(32)^{-\frac{1}{5}}}$

$$\begin{aligned}
 &= \frac{1}{(6^3)^{-\frac{2}{3}}} + \frac{1}{(4^4)^{-\frac{3}{4}}} + \frac{1}{(2^5)^{-\frac{1}{5}}} \\
 &= \frac{1}{6^{3 \times \left(-\frac{2}{3}\right)}} + \frac{1}{4^{4 \times \left(-\frac{3}{4}\right)}} + \frac{1}{2^{5 \times \left(-\frac{1}{5}\right)}} = \frac{1}{6^{-2}} + \frac{1}{4^{-3}} + \frac{1}{2^{-1}} \\
 &= (6^2 + 4^3 + 2^1) = (36 + 64 + 2) = 102.
 \end{aligned}$$

33. $(2.4 \times 10^3) \div (8 \times 10^{-2}) = \frac{2.4 \times 10^3}{8 \times 10^{-2}} = \frac{24 \times 10^2}{8 \times 10^{-2}} = (3 \times 10^4).$

34. $\left(\frac{1}{216}\right)^{-\frac{2}{3}} + \left(\frac{1}{27}\right)^{-\frac{4}{3}} = (216)^{\frac{2}{3}} + (27)^{\frac{4}{3}} = \frac{(216)^{\frac{2}{3}}}{(27)^{\frac{4}{3}}} = \frac{(6^3)^{\frac{2}{3}}}{(3^3)^{\frac{4}{3}}}$

$$\begin{aligned}
 &= \frac{6^{\left(3 \times \frac{2}{3}\right)}}{3^{\left(3 \times \frac{4}{3}\right)}} = \frac{6^2}{3^4} = \frac{36}{81} = \frac{4}{9}.
 \end{aligned}$$

35. $(48)^{-\frac{2}{7}} \times (16)^{-\frac{5}{7}} \times (3)^{-\frac{5}{7}} = (16 \times 3)^{-\frac{2}{7}} \times (16)^{-\frac{5}{7}} \times (3)^{-\frac{5}{7}}$

$$\begin{aligned}
 &= (16)^{-\frac{2}{7}} \times (3)^{-\frac{2}{7}} \times (16)^{-\frac{5}{7}} \times (3)^{-\frac{5}{7}} \\
 &= (16)^{\left(-\frac{2}{7} - \frac{5}{7}\right)} \times (3)^{\left(-\frac{2}{7} - \frac{5}{7}\right)} \\
 &= (16)^{\left(-\frac{7}{7}\right)} \times (3)^{\left(-\frac{7}{7}\right)} = (16)^{-1} \times (3)^{-1} \\
 &= \frac{1}{16} \times \frac{1}{3} = \frac{1}{48}.
 \end{aligned}$$

36. $10^{-8x} = (10^x)^{-8} = \left(\frac{1}{2}\right)^{-8} = 2^8 = 256.$

37. $\left(\frac{3}{5}\right)^3 \left(\frac{3}{5}\right)^{-6} = \left(\frac{3}{5}\right)^{2x-1} \Rightarrow \left(\frac{3}{5}\right)^{(3-6)} = \left(\frac{3}{5}\right)^{2x-1}$

$$\Rightarrow \left(\frac{3}{5}\right)^{-3} = \left(\frac{3}{5}\right)^{2x-1}$$

$$\Rightarrow 2x - 1 = -3 \Rightarrow 2x = -2 \Rightarrow x = -1.$$

38. $49 \times 49 \times 49 \times 49 = 7^2 \times 7^2 \times 7^2 \times 7^2 = 7^{(2+2+2+2)} = 7^8.$

$\therefore$ Required number = 8.

39. $8^{-25} - 8^{-26} = \left(\frac{1}{8^{25}} - \frac{1}{8^{26}}\right) = \frac{(8-1)}{8^{26}} = 7 \times 8^{-26}.$

40. $(64)^{-\frac{1}{2}} - (-32)^{-\frac{4}{5}} = (8^2)^{-\frac{1}{2}} - [(-2)^5]^{-\frac{4}{5}}$

$$\begin{aligned}
 &= 8^{\left[2 \times \left(-\frac{1}{2}\right)\right]} - (-2)^{\left[5 \times \left(-\frac{4}{5}\right)\right]} \\
 &= 8^{-1} - (-2)^{-4} = \frac{1}{8} - \frac{1}{(-2)^4} \\
 &= \left(\frac{1}{8} - \frac{1}{16}\right) = \frac{1}{16}.
 \end{aligned}$$

$$41. \left(\frac{a}{b}\right)^{x-1} = \left(\frac{b}{a}\right)^{x-3} \Leftrightarrow \left(\frac{a}{b}\right)^{x-1} = \left(\frac{a}{b}\right)^{-(x-3)} = \left(\frac{a}{b}\right)^{(3-x)}$$

$$\Leftrightarrow x-1 = 3-x \Leftrightarrow 2x = 4 \Leftrightarrow x = 2.$$

$$42. 2^{2n-1} = \frac{1}{8^{n-3}} \Leftrightarrow 2^{2n-1} = \frac{1}{(2^3)^{n-3}}$$

$$= \frac{1}{2^{3(n-3)}} = \frac{1}{2^{(9-3n)}} = 2^{(9-3n)}$$

$$\Leftrightarrow 2n-1 = 9-3n \Leftrightarrow 5n = 10$$

$$\Leftrightarrow n = 2.$$

$$43. 5^n = 3125 = 5^5 \Rightarrow a = 5. \Rightarrow 5^{(a-3)} = 5^{(5-3)} = 5^2 = 25.$$

$$44. 5\sqrt{5} \times 5^3 + 5^{-\frac{3}{2}} = 5^{a+2} \Leftrightarrow \frac{5 \times 5^{\frac{1}{2}} \times 5^3}{5^{-\frac{3}{2}}} = 5^{a+2}$$

$$\Leftrightarrow 5^{\left(1+\frac{1}{2}+3+\frac{3}{2}\right)} = 5^{a+2} \Leftrightarrow 5^6 = 5^{a+2}$$

$$\Leftrightarrow a+2 = 6 \Leftrightarrow a = 4.$$

$$45. \sqrt{2^n} = 64 \Leftrightarrow (2^n)^{\frac{1}{2}} = 2^6 \Leftrightarrow 2^{\frac{n}{2}} = 2^6 \Leftrightarrow \frac{n}{2} = 6$$

$$\Leftrightarrow n = 12.$$

$$46. (\sqrt{3})^5 \times 9^2 = 3^n \times 3\sqrt{3} \Leftrightarrow \left(\frac{1}{3^2}\right)^5 \times (3^2)^2 = 3^n \times 3 \times 3^{\frac{1}{2}}$$

$$\Leftrightarrow 3^{\left(\frac{1}{2} \times 5\right)} \times 3^{(2 \times 2)} = 3^{\left(n+1+\frac{1}{2}\right)}$$

$$\Leftrightarrow 3^{\left(\frac{5}{2}+4\right)} = 3^{\left(n+\frac{3}{2}\right)} \Leftrightarrow n+\frac{3}{2} = \frac{13}{2}$$

$$\Leftrightarrow n = \left(\frac{13}{2} - \frac{3}{2}\right) = \frac{10}{2} = 5.$$

$$47. \frac{9^n \times 3^5 \times (27)^3}{3 \times (81)^4} = 27 \Leftrightarrow \frac{(3^2)^n \times 3^5 \times (3^3)^3}{3 \times (3^4)^4} = 3^3$$

$$\Leftrightarrow \frac{3^{2n} \times 3^5 \times 3^{(3 \times 3)}}{3 \times 3^{(4 \times 4)}} = 3^3$$

$$\Leftrightarrow \frac{3^{2n+5+9}}{3 \times 3^{16}} = 3^3 \Leftrightarrow \frac{3^{2n+14}}{3^{17}} = 3^3$$

$$\Leftrightarrow 3^{(2n+14-17)} = 3^3$$

$$\Leftrightarrow 3^{2n-3} = 3^3 \Leftrightarrow 2n-3 = 3 \Leftrightarrow 2n = 6 \Leftrightarrow n = 3.$$

$$48. \left(\frac{9}{4}\right)^x \cdot \left(\frac{8}{27}\right)^{x-1} = \frac{2}{3} \Leftrightarrow \frac{9^x}{4^x} \times \frac{8^{(x-1)}}{27^{(x-1)}} = \frac{2}{3}$$

$$\Leftrightarrow \frac{(3^2)^x \times (2^3)^{(x-1)}}{(2^2)^x \times (3^3)^{(x-1)}} = \frac{2}{3} \Leftrightarrow \frac{3^{2x} \times 2^{3(x-1)}}{2^{2x} \times 3^{3(x-1)}} = \frac{2}{3}$$

$$\Leftrightarrow \frac{2^{(3x-3-2x)}}{3^{(3x-3-2x)}} = \frac{2}{3} \Leftrightarrow \frac{2^{(x-3)}}{3^{(x-3)}} = \frac{2}{3} \Leftrightarrow \left(\frac{2}{3}\right)^{(x-3)} = \left(\frac{2}{3}\right)^1$$

$$\Leftrightarrow x-3 = 1 \Leftrightarrow x = 4.$$

$$49. 2^x = \sqrt[3]{32} \Leftrightarrow 2^x = (32)^{\frac{1}{3}} = (2^5)^{\frac{1}{3}} = 2^{\frac{5}{3}} \Leftrightarrow x = \frac{5}{3}.$$

$$50. 2^x \times 8^{\frac{1}{5}} = 2^{\frac{1}{5}} \Leftrightarrow 2^x \times (2^3)^{\frac{1}{5}} = 2^{\frac{1}{5}} \Leftrightarrow 2^x \times 2^{\frac{3}{5}} = 2^{\frac{1}{5}}$$

$$\Leftrightarrow 2^{\left(x+\frac{3}{5}\right)} = 2^{\frac{1}{5}}$$

$$\Leftrightarrow x+\frac{3}{5} = \frac{1}{5} \Leftrightarrow x = \left(\frac{1}{5} - \frac{3}{5}\right) = -\frac{2}{5}.$$

$$51. 5^{(x+3)} = 25^{(3x-4)} \Leftrightarrow 5^{(x+3)} = (5^2)^{(3x-4)} \Leftrightarrow 5^{(x+3)} = 5^{2(3x-4)}$$

$$= 5^{(6x-8)} \Leftrightarrow 5^{(x+3)} = 5^{(6x-8)}$$

$$\Leftrightarrow x+3 = 6x-8 \Leftrightarrow 5x = 11$$

$$\Leftrightarrow x = \frac{11}{5}.$$

$$52. \frac{2^{n+4} - 2(2^n)}{2(2^{n+3})} = \frac{2^{n+4} - 2^{n+1}}{2^{n+4}} = \frac{2^{n+4}}{2^{n+4}} - \frac{2^{n+1}}{2^{n+4}}$$

$$= 1 - 2^{n+1-(n+4)} = 1 - 2^{-3} = 1 - \frac{1}{8} = \frac{7}{8}.$$

$$53. \left[\sqrt[3]{6\sqrt{a^9}}\right]^4 \left[\sqrt[6]{3\sqrt{a^9}}\right]^4 = \left[\left\{\left(a^9\right)^{\frac{1}{6}}\right\}^{\frac{1}{3}}\right]^4 \cdot \left[\left\{\left(a^9\right)^{\frac{1}{3}}\right\}^{\frac{1}{6}}\right]^4$$

$$= a^{\left(9 \times \frac{1}{6} \times \frac{1}{3} \times 4\right)} \cdot a^{\left(9 \times \frac{1}{3} \times \frac{1}{6} \times 4\right)}$$

$$= a^2 \cdot a^2 = a^4.$$

$$54. (256)^{0.16} \times (256)^{0.09} = (256)^{(0.16+0.09)} = (256)^{0.25}$$

$$= (256)^{\left(\frac{25}{100}\right)}$$

$$= (256)^{\frac{1}{4}} = (4^4)^{\frac{1}{4}} = 4^{\left(4 \times \frac{1}{4}\right)} = 4^1 = 4.$$

$$55. (0.04)^{-1.5} = \left(\frac{4}{100}\right)^{-1.5} = \left(\frac{1}{25}\right)^{-\frac{3}{2}} = (25)^{\frac{3}{2}} = (5^2)^{\frac{3}{2}}$$

$$= 5^{\left(2 \times \frac{3}{2}\right)} = 5^3 = 125.$$

$$56. \text{ Let } (17)^{3.5} \times (17)^x = 17^8. \text{ Then, } (17)^{(3.5+x)} = (17)^8.$$

$$\therefore 3.5+x = 8 \Leftrightarrow x = (8-3.5)$$

$$\Leftrightarrow x = 4.5.$$

$$57. \text{ Let } 6^{1.2} \times 36^x \times 30^{2.4} \times 25^{1.3} = 30^5.$$

$$\text{Then, } 6^{1.2} \times (6^2)^x \times (6 \times 5)^{2.4} \times (5^2)^{1.3} = 30^5$$

$$\Leftrightarrow 6^{1.2} \times 6^{2x} \times 6^{2.4} \times 5^{2.4} \times 5^{2.6} = (6 \times 5)^5$$

$$\Leftrightarrow 6^{(1.2+2x+2.4)} \times 5^{(2.4+2.6)} = 6^5 \times 5^5$$

$$\Leftrightarrow 6^{(3.6+2x)} \times 5^5 = 6^5 \times 5^5 \Leftrightarrow 3.6+2x = 5 \Leftrightarrow 2x = 1.4$$

$$\Leftrightarrow x = 0.7.$$

$$58. \text{ Let } 2^{3.6} \times 4^{3.6} \times 4^{3.6} \times 32^{2.3} = 32^x.$$

$$\text{Then, } 2^{3.6} \times (2^2)^{3.6} \times (2^2)^{3.6} \times (2^5)^{2.3} = (2^5)^x$$

$$\Leftrightarrow 2^{3.6} \times 2^{(2 \times 3.6)} \times 2^{(2 \times 3.6)} \times (2^5)^{2.3} = (2^5)^x$$

$$\Leftrightarrow 2^{(3.6+7.2+7.2)} \times (2^5)^{2.3} = (2^5)^x \Leftrightarrow 2^{18} \times (2^5)^{2.3} = (2^5)^x$$

$$\Leftrightarrow (2^5)^{3.6} \times (2^5)^{2.3} = (2^5)^x \Leftrightarrow (2^5)^{(3.6+2.3)} = (2^5)^x$$

$$\Leftrightarrow (2^5)^{5.9} = (2^5)^x \Leftrightarrow x = 5.9.$$

$$59. \text{ Let } 3^{3.5} \times (21)^2 \times (42)^{2.5} \div 2^{2.5} \times 7^{3.5} = 21^x.$$

$$\text{Then, } 3^{3.5} \times 7^{3.5} \times (21)^2 \times (21 \times 2)^{2.5} \div 2^{2.5} = 21^x$$

$$\Leftrightarrow (21)^x = (3 \times 7)^{3.5} \times (21)^2 \times (21)^{2.5} \times 2^{2.5} \div 2^{2.5}$$

- $\Leftrightarrow (21)^x = (21)^{3.5} \times (21)^{(2+2.5)} \Leftrightarrow (21)^x = (21)^{3.5} \times (21)^{(4.5)}$
 $\Leftrightarrow (21)^x = (21)^{(3.5+4.5)} = (21)^8$
 $\Leftrightarrow x = 8$
60. $8^{0.4} \times 4^{1.6} \times 2^{1.6} = (2^3)^{0.4} \times (2^2)^{1.6} \times 2^{1.6}$
 $= 2^{(3 \times 0.4)} \times 2^{(2 \times 1.6)} \times 2^{1.6}$
 $= 2^{1.2} \times 2^{3.2} \times 2^{1.6} = 2^{(1.2+3.2+1.6)} = 2^6 = 64$.
61. Let $8^7 \times 2^6 \div 8^{2.4} = 8^x$.
Then, $8^7 \times (2^3)^2 \div 8^{2.4} = 8^x$
 $\Leftrightarrow 8^7 \times 8^2 \div 8^{2.4} = 8^x$
 $\Leftrightarrow 8^x = 8^{(7+2-2.4)} \Leftrightarrow 8^x = 8^{6.6} \Leftrightarrow x = 6.6$.
62. Let $25^{2.7} \times 5^{4.2} \div 5^{5.4} = 25^x$.
Then, $(25)^{2.7} \times 5^{(4.2-5.4)} = 25^x$
 $\Leftrightarrow (25)^{2.7} \times 5^{(-1.2)} = 25^x \Leftrightarrow (25)^{2.7} \times \frac{1}{5^{1.2}} = 25^x$
 $\Leftrightarrow \frac{(25)^{2.7}}{(5^2)^{0.6}} = 25^x \Leftrightarrow \frac{(25)^{2.7}}{(25)^{0.6}} = 25^x$
 $\Leftrightarrow 25^x = 25^{(2.7-0.6)} = 25^{2.1} \Leftrightarrow x = 2.1$
63. Let $8^{2.4} \times 2^{3.7} \div (16)^{1.3} = 2^x$.
Then, $(2^3)^{2.4} \times 2^{3.7} \div (2^4)^{1.3} = 2^x$
 $\Leftrightarrow 2^{(3 \times 2.4)} \times 2^{3.7} \div 2^{(4 \times 1.3)} = 2^x$
 $\Leftrightarrow 2^{7.2} \times 2^{3.7} \div 2^{5.2} = 2^x$
 $\Leftrightarrow 2^x = 2^{(7.2+3.7-5.2)} = 2^{5.7} \Leftrightarrow x = 5.7$.
64. Let $(0.04)^2 \div (0.008) \times (0.2)^6 = (0.2)^x$.
Then, $(0.2)^x = [(0.2)^2]^2 \div (0.2)^3 \times (0.2)^6$
 $\Leftrightarrow (0.2)^x = (0.2)^{(2 \times 2)} \div (0.2)^3 \times (0.2)^6$
 $\Leftrightarrow (0.2)^x = (0.2)^4 \div (0.2)^3 \times (0.2)^6 = (0.2)^{(4-3+6)} = (0.2)^7$
 $\Leftrightarrow x = 7$.
65. $(18)^{3.5} \div (27)^{3.5} \times 6^{3.5} = 2^x$
 $\Leftrightarrow (18)^{3.5} \times \frac{1}{(27)^{3.5}} \times 6^{3.5} = 2^x$
 $\Leftrightarrow (3^2 \times 2)^{3.5} \times \frac{1}{(3^3)^{3.5}} \times (2 \times 3)^{3.5} = 2^x$
 $\Leftrightarrow 3^{(2 \times 3.5)} \times 2^{3.5} \times \frac{1}{3^{(3 \times 3.5)}} \times 2^{3.5} \times 3^{3.5} = 2^x$
 $\Leftrightarrow 3^7 \times 2^{3.5} \times \frac{1}{3^{10.5}} \times 2^{3.5} \times 3^{3.5} = 2^x$
 $\Leftrightarrow \frac{3^{(7+3.5)}}{3^{10.5}} \times 2^{(3.5+3.5)} \Leftrightarrow 2^7 = 2^x \Leftrightarrow x = 7$.
66. Let $(25)^{7.5} \times (5)^{2.5} \div (125)^{1.5} = 5^x$.
Then, $\frac{(5^2)^{7.5} \times (5)^{2.5}}{(5^3)^{1.5}} = 5^x \Leftrightarrow \frac{5^{(2 \times 7.5)} \times 5^{2.5}}{5^{(3 \times 1.5)}} = 5^x$
 $\Leftrightarrow 1 \frac{5^{15} \times 5^{2.5}}{5^{4.5}} = 5^x$
 $\Leftrightarrow 5^x = 5^{(15+2.5-4.5)} = 5^{13} \Leftrightarrow x = 13$.
67. $\frac{(243)^{0.13} \times (243)^{0.07}}{7^{0.25} \times (49)^{0.075} \times (343)^{0.2}} = \frac{(243)^{(0.13+0.07)}}{7^{0.25} \times (7^2)^{0.075} \times (7^3)^{0.2}}$
 $= \frac{(243)^{0.2}}{7^{0.25} \times 7^{(2 \times 0.075)} \times 7^{(3 \times 0.2)}}$

- $$= \frac{(3^5)^{0.2}}{7^{0.25} \times 7^{0.15} \times 7^{0.6}} = \frac{3^{(5 \times 0.2)}}{7^{(0.25+0.15+0.6)}} = \frac{3^1}{7^1} = \frac{3}{7}$$
68. $(64x^3 + 27a^{-3})^{-\frac{2}{3}} = \left(\frac{64x^3}{27a^{-3}} \right)^{-\frac{2}{3}} = \left(\frac{4^3 \cdot x^3}{3^3 \cdot a^{-3}} \right)^{-\frac{2}{3}}$
 $= \left(\frac{4^3 \cdot x^3 \cdot a^3}{3^3} \right)^{-\frac{2}{3}}$
 $= \frac{\{(4ax)^3\}^{-\frac{2}{3}}}{3^{3 \times (-\frac{2}{3})}} = \frac{(4ax)^{3 \times (-\frac{2}{3})}}{3^{-2}} = \frac{(4ax)^{-2}}{3^{-2}}$
 $= \frac{3^2}{(4ax)^2} = \frac{9}{16a^2x^2}$.
69. $2^{n+4} - 2^{n+2} = 3 \Leftrightarrow 2^{n+2}(2^2 - 1) = 3$
 $\Leftrightarrow 2^{n+2} = 1 = 2^0$
 $\Leftrightarrow n + 2 = 0 \Leftrightarrow n = -2$.
70. $2^{n-1} + 2^{n+1} = 320 \Leftrightarrow 2^{n-1}(1 + 2^2) = 320$
 $\Leftrightarrow 5 \times 2^{n-1} = 320$
 $\Leftrightarrow 2^{n-1} = \frac{320}{5} = 64 = 2^6$
 $\Leftrightarrow n - 1 = 6 \Leftrightarrow n = 7$.
71. $3^x - 3^{x-1} = 18 \Leftrightarrow 3^{x-1}(3 - 1) = 18$
 $\Leftrightarrow 3^{x-1} = 9 = 3^2 \Leftrightarrow x - 1 = 2 \Leftrightarrow x = 3$.
 $\therefore x^x = 3^3 = 27$.
72. $\frac{2^{n+4} - 2 \times 2^n}{2 \times 2^{n+3}} + 2^{-3} = \frac{2^{n+4} - 2^{n+1}}{2^{n+4}} + \frac{1}{2^3}$
 $= \frac{2^{n+1}(2^3 - 1)}{2^{n+4}} + \frac{1}{2^3}$
 $= \frac{2^{n+1} \times 7}{2^{n+1} \times 2^3} + \frac{1}{2^3} = \left(\frac{7}{8} + \frac{1}{8} \right) = \frac{8}{8} = 1$.
73. $\frac{2^{3x+4} + 8^{x+1}}{8^{x+1} - 2^{3x+2}} = \frac{2^{3x+4} + (2^3)^{x+1}}{(2^3)^{x+1} - 2^{3x+2}} = \frac{2^{3x+4} + 2^{3(x+1)}}{2^{3(x+1)} - 2^{3x+2}}$
 $= \frac{2^{3x+4} + 2^{3x+3}}{2^{3x+3} - 2^{3x+2}} = \frac{2^{3x+3}(2+1)}{2^{3x+2}(2-1)}$
 $= 3.2^{(3x+3)-(3x+2)} = 3.2^1$
 $= 3 \times 2 = 6$.
74. $\frac{2^{n-1} - 2^n}{2^{n+4} + 2^{n+1}} = \frac{2^{n-1}(1-2)}{2^{n+1}(2^3+1)} = \left(-\frac{1}{9} \right) \cdot 2^{(n-1)-(n+1)}$
 $= \left(-\frac{1}{9} \right) \cdot 2^{-2} = \left(-\frac{1}{9} \right) \times \frac{1}{2^2} = \left(-\frac{1}{9} \right) \times \frac{1}{4}$
 $= -\frac{1}{36}$.
75. $\left(\sqrt{x} - \frac{1}{\sqrt{x}} \right)^2 = x + \frac{1}{x} - 2 = (5 + 2\sqrt{6}) + \frac{1}{(5 + 2\sqrt{6})} - 2$
 $= (5 + 2\sqrt{6}) + \frac{1}{(5 + 2\sqrt{6})} \times \frac{(5 - 2\sqrt{6})}{(5 - 2\sqrt{6})} - 2$
 $= (5 + 2\sqrt{6}) + (5 - 2\sqrt{6}) - 2 = 8$.

$$\therefore \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right) = \sqrt{8} = 2\sqrt{2}.$$

$$\begin{aligned} 76. \quad 4 + \sqrt{7} &= \frac{7}{2} + \frac{1}{2} + 2 \times \frac{\sqrt{7}}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \\ &= \left(\frac{\sqrt{7}}{\sqrt{2}} \right)^2 + \left(\frac{1}{\sqrt{2}} \right)^2 + 2 \times \frac{\sqrt{7}}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \\ &= \left(\frac{\sqrt{7}}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right)^2 = \frac{1}{2} (\sqrt{7} + 1)^2. \end{aligned}$$

$$\begin{aligned} 77. \quad \sqrt{8 - 2\sqrt{15}} &= \sqrt{5 + 3 - 2 \times \sqrt{5} \times \sqrt{3}} \\ &= \sqrt{(\sqrt{5})^2 + (\sqrt{3})^2 - 2 \times \sqrt{5} \times \sqrt{3}} \\ &= \sqrt{(\sqrt{5} - \sqrt{3})^2} = (\sqrt{5} - \sqrt{3}). \end{aligned}$$

$$\begin{aligned} 78. \quad \sqrt{6 - 4\sqrt{3}} + \sqrt{16 - 8\sqrt{3}} &= \sqrt{6 - 4\sqrt{3}} + \sqrt{12 + 4 - 8\sqrt{3}} \\ &= \sqrt{6 - 4\sqrt{3}} + \sqrt{(2\sqrt{3})^2 + (2)^2 - 2 \times 2\sqrt{3} \times 2} \\ &= \sqrt{6 - 4\sqrt{3}} + \sqrt{(2\sqrt{3} - 2)^2} = \sqrt{6 - 4\sqrt{3}} + 2\sqrt{3} - 2 \\ &= \sqrt{(\sqrt{3})^2 + (1)^2 - 2 \times \sqrt{3} \times 1} = \sqrt{(\sqrt{3} - 1)^2} = \sqrt{3} - 1. \end{aligned}$$

$$\begin{aligned} 79. \quad \frac{1}{\sqrt{12 - \sqrt{140}}} - \frac{1}{\sqrt{8 - \sqrt{60}}} - \frac{2}{\sqrt{10 + \sqrt{84}}} \\ &= \frac{1}{\sqrt{12 - \sqrt{4 \times 35}}} - \frac{1}{\sqrt{8 - \sqrt{4 \times 15}}} - \frac{2}{\sqrt{10 + \sqrt{4 \times 21}}} \\ &= \frac{1}{\sqrt{12 - 2\sqrt{35}}} - \frac{1}{\sqrt{8 - 2\sqrt{15}}} - \frac{2}{\sqrt{10 + 2\sqrt{21}}} \\ &= \frac{1}{\sqrt{7 + 5 - 2\sqrt{35}}} - \frac{1}{\sqrt{5 + 3 - 2\sqrt{15}}} - \frac{2}{\sqrt{7 + 3 + 2\sqrt{21}}} \\ &= \frac{1}{\sqrt{(\sqrt{7})^2 + (\sqrt{5})^2 - 2 \times \sqrt{7} \times \sqrt{5}}} - \frac{1}{\sqrt{(\sqrt{5})^2 + (\sqrt{3})^2 - 2 \times \sqrt{5} \times \sqrt{3}}} \\ &\quad - \frac{2}{\sqrt{(\sqrt{7})^2 + (\sqrt{3})^2 + 2 \times \sqrt{7} \times \sqrt{3}}} \\ &= \frac{1}{\sqrt{(\sqrt{7} - \sqrt{5})^2}} - \frac{1}{\sqrt{(\sqrt{5} - \sqrt{3})^2}} - \frac{2}{\sqrt{(\sqrt{7} + \sqrt{3})^2}} \\ &= \frac{1}{(\sqrt{7} - \sqrt{5})} - \frac{1}{(\sqrt{5} - \sqrt{3})} - \frac{2}{(\sqrt{7} + \sqrt{3})} \\ &= \frac{1}{\sqrt{7} - \sqrt{5}} \times \frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} + \sqrt{5}} - \frac{1}{\sqrt{5} - \sqrt{3}} \times \frac{\sqrt{5} + \sqrt{3}}{\sqrt{5} + \sqrt{3}} \\ &\quad - \frac{2}{\sqrt{7} + \sqrt{3}} \times \frac{\sqrt{7} - \sqrt{3}}{\sqrt{7} - \sqrt{3}} \\ &= \frac{\sqrt{7} + \sqrt{5}}{7 - 5} - \frac{\sqrt{5} + \sqrt{3}}{5 - 3} - \frac{2(\sqrt{7} - \sqrt{3})}{7 - 3} \\ &= \frac{(\sqrt{7} + \sqrt{5})}{2} - \frac{(\sqrt{5} + \sqrt{3})}{2} - \frac{(\sqrt{7} - \sqrt{3})}{2} \end{aligned}$$

$$= \frac{\sqrt{7} + \sqrt{5} - \sqrt{5} - \sqrt{3} - \sqrt{7} + \sqrt{3}}{2} = 0.$$

$$\begin{aligned} 80. \quad \sqrt{4 + \sqrt{15}} &= \sqrt{\frac{5}{2} + \frac{3}{2} + 2 \times \frac{\sqrt{5}}{\sqrt{2}} \times \frac{\sqrt{3}}{\sqrt{2}}} \\ &= \sqrt{\left(\frac{\sqrt{5}}{\sqrt{2}} \right)^2 + \left(\frac{\sqrt{3}}{\sqrt{2}} \right)^2 + 2 \times \frac{\sqrt{5}}{\sqrt{2}} \times \frac{\sqrt{3}}{\sqrt{2}}} \\ &= \sqrt{\left(\frac{\sqrt{5}}{\sqrt{2}} + \frac{\sqrt{3}}{\sqrt{2}} \right)^2} = \frac{\sqrt{5}}{\sqrt{2}} + \frac{\sqrt{3}}{\sqrt{2}}. \end{aligned}$$

$$\text{Similarly, } \sqrt{4 - \sqrt{15}} = \frac{\sqrt{5}}{\sqrt{2}} - \frac{\sqrt{3}}{\sqrt{2}}.$$

$$\begin{aligned} \sqrt{12 - 4\sqrt{3}} &= \sqrt{10 + 2 - 2 \times \sqrt{10} \times \sqrt{2}} \\ &= \sqrt{(\sqrt{10})^2 + (\sqrt{2})^2 - 2 \times \sqrt{10} \times \sqrt{2}} \\ &= \sqrt{(\sqrt{10} - \sqrt{2})^2} = (\sqrt{10} - \sqrt{2}). \end{aligned}$$

$\therefore$ Given expression

$$\begin{aligned} &= \left(\frac{\sqrt{5}}{\sqrt{2}} + \frac{\sqrt{3}}{\sqrt{2}} \right) + \left(\frac{\sqrt{5}}{\sqrt{2}} - \frac{\sqrt{3}}{\sqrt{2}} \right) - (\sqrt{10} - \sqrt{2}) \\ &= \frac{2\sqrt{5}}{\sqrt{2}} - \sqrt{10} + \sqrt{2} = \sqrt{10} - \sqrt{10} + \sqrt{2} = \sqrt{2}, \end{aligned}$$

which is an irrational number.

$$81. \quad \text{Let } X = \frac{\sqrt{\sqrt{5} + 2} + \sqrt{\sqrt{5} - 2}}{\sqrt{\sqrt{5} + 1}}.$$

$$\begin{aligned} \text{Then } X^2 &= \frac{(\sqrt{\sqrt{5} + 2} + \sqrt{\sqrt{5} - 2})^2}{(\sqrt{\sqrt{5} + 1})^2} \\ &= \frac{(\sqrt{5} + 2) + (\sqrt{5} - 2) + 2\sqrt{(\sqrt{5} + 2)(\sqrt{5} - 2)}}{(\sqrt{5} + 1)} \\ &= \frac{2\sqrt{5} + 2\sqrt{(\sqrt{5})^2 - (2)^2}}{\sqrt{5} + 1} = \frac{2\sqrt{5} + 2}{\sqrt{5} + 1} \\ &= \frac{2(\sqrt{5} + 1)}{(\sqrt{5} + 1)} = 2 \end{aligned}$$

$$\Rightarrow X = \sqrt{2}.$$

$$\begin{aligned} \therefore N &= \sqrt{2} - \sqrt{3 - 2\sqrt{2}} = \sqrt{2} - \sqrt{(\sqrt{2})^2 + 1^2 - 2 \times \sqrt{2} \times 1} \\ &= \sqrt{2} - \sqrt{(\sqrt{2} - 1)^2} = \sqrt{2} - (\sqrt{2} - 1) = 1. \end{aligned}$$

$$\begin{aligned} 82. \quad x^z = y^2 &\Leftrightarrow (10^{0.48})^z = (10^{0.70})^2 \\ &\Leftrightarrow 10^{(0.48z)} = 10^{(2 \times 0.70)} = 10^{1.40} \\ &\Leftrightarrow 0.48z = 1.40 \\ &\Leftrightarrow z = \frac{140}{48} = \frac{35}{12} = 2.9 \text{ (approx.)} \end{aligned}$$

83. We know that $11^2 = 121$.

Putting $m = 11$ and $n = 2$, we get :

$$(m - 1)^{n+1} = (11 - 1)^{(2+1)} = 10^3 = 1000.$$

$$\begin{aligned}
 84. \quad & (216)^{\frac{3}{5}} \times (2500)^{\frac{2}{5}} \times (300)^{\frac{1}{5}} = (3^3 \times 2^3)^{\frac{3}{5}} \times (5^4 \times 2^2)^{\frac{2}{5}} \\
 & \times (5^2 \times 2^2 \times 3)^{\frac{1}{5}} \\
 & = 3^{\left(3 \times \frac{3}{5}\right)} \times 2^{\left(3 \times \frac{3}{5}\right)} \times 5^{\left(4 \times \frac{2}{5}\right)} \times 2^{\left(2 \times \frac{2}{5}\right)} \times 5^{\left(2 \times \frac{1}{5}\right)} \\
 & \times 2^{\left(2 \times \frac{1}{5}\right)} \times 3^{\frac{1}{5}} \\
 & = 3^{\frac{9}{5}} \times 2^{\frac{9}{5}} \times 5^{\frac{8}{5}} \times 2^{\frac{4}{5}} \times 5^{\frac{2}{5}} \times 2^{\frac{2}{5}} \times 3^{\frac{1}{5}} \\
 & = 3^{\left(\frac{9}{5} + \frac{1}{5}\right)} \times 2^{\left(\frac{9}{5} + \frac{4}{5} + \frac{2}{5}\right)} \times 5^{\left(\frac{8}{5} + \frac{2}{5}\right)} = 3^2 \times 2^3 \times 5^2.
 \end{aligned}$$

Hence, the number of prime factors = $(2 + 3 + 2) = 7$.

$$\begin{aligned}
 85. \quad & \frac{6^{12} \times (35)^{28} \times (15)^{16}}{(14)^{12} \times (21)^{11}} = \frac{(2 \times 3)^{12} \times (5 \times 7)^{28} \times (3 \times 5)^{16}}{(2 \times 7)^{12} \times (3 \times 7)^{11}} \\
 & = \frac{2^{12} \times 3^{12} \times 5^{28} \times 7^{28} \times 3^{16} \times 5^{16}}{2^{12} \times 7^{12} \times 3^{11} \times 7^{11}} \\
 & = 2^{(12-12)} \times 3^{(12+16-11)} \times 5^{(28+16)} \times 7^{(28-12-11)} \\
 & = 2^0 \times 3^{17} \times 5^{44} \times 7^5 = 3^{17} \times 5^{44} \times 7^5.
 \end{aligned}$$

Number of prime factors = $17 + 44 + 5 = 66$.

$$\begin{aligned}
 86. \quad & 1 + (3 + 1) (3^2 + 1) (3^4 + 1) (3^8 + 1) (3^{16} + 1) (3^{32} + 1) \\
 & = 1 + \frac{1}{2} [(3 - 1) (3 + 1) (3^2 + 1) (3^4 + 1) (3^8 + 1) \\
 & \quad (3^{16} + 1) (3^{32} + 1)] \\
 & = 1 + \frac{1}{2} [(3^2 - 1) (3^2 + 1) (3^4 + 1) (3^8 + 1) \\
 & \quad (3^{16} + 1) (3^{32} + 1)] \\
 & = 1 + \frac{1}{2} [(3^4 - 1) (3^4 + 1) (3^8 + 1) (3^{16} + 1) (3^{32} + 1)] \\
 & = 1 + \frac{1}{2} [(3^8 - 1) (3^8 + 1) (3^{16} + 1) (3^{32} + 1)] \\
 & = 1 + \frac{1}{2} [(3^{16} - 1) (3^{16} + 1) (3^{32} + 1)] \\
 & = 1 + \frac{1}{2} [(3^{32} - 1) (3^{32} + 1)] \\
 & = 1 + \frac{1}{2} (3^{64} - 1) = \frac{2 + 3^{64} - 1}{2} = \frac{3^{64} + 1}{2}.
 \end{aligned}$$

$$\begin{aligned}
 87. \quad & \frac{1}{1 + a^{(n-m)}} + \frac{1}{1 + a^{(m-n)}} = \frac{1}{\left(1 + \frac{a^n}{a^m}\right)} + \frac{1}{\left(1 + \frac{a^m}{a^n}\right)} \\
 & = \frac{a^m}{(a^m + a^n)} + \frac{a^n}{(a^m + a^n)} = \frac{(a^m + a^n)}{(a^m + a^n)} = 1.
 \end{aligned}$$

$$\begin{aligned}
 88. \quad & (x^a)^{a^2 - bc} (x^b)^{b^2 - ca} (x^c)^{c^2 - ab} \\
 & = x^{[a(a^2 - bc)]} \cdot x^{[b(b^2 - ca)]} \cdot x^{[c(c^2 - ab)]} \\
 & = x^{(a^3 - abc)} \cdot x^{(b^3 - abc)} \cdot x^{(c^3 - abc)} \\
 & = x^{(a^3 - abc + b^3 - abc + c^3 - abc)} = x^{(a^3 + b^3 + c^3 - 3abc)} \\
 & = x^{(3abc - 3abc)} = x^0 = 1. \\
 & [\therefore \text{ If } a + b + c = 0, a^3 + b^3 + c^3 = 3abc]
 \end{aligned}$$

$$\begin{aligned}
 89. \quad & \text{Given Exp.} = \frac{1}{\left(1 + \frac{x^b}{x^a} + \frac{x^c}{x^a}\right)} + \frac{1}{\left(1 + \frac{x^a}{x^b} + \frac{x^c}{x^b}\right)} \\
 & + \frac{1}{\left(1 + \frac{x^b}{x^c} + \frac{x^a}{x^c}\right)} \\
 & = \frac{x^a}{(x^a + x^b + x^c)} + \frac{x^b}{(x^a + x^b + x^c)} \\
 & + \frac{x^c}{(x^a + x^b + x^c)} \\
 & = \frac{(x^a + x^b + x^c)}{(x^a + x^b + x^c)} = 1.
 \end{aligned}$$

$$\begin{aligned}
 90. \quad & \text{Given Exp.} \\
 & = x^{(b-c)} (b+c-a) \cdot x^{(c-a)} (c+a-b) \cdot x^{(a-b)} (a+b-c) \\
 & = x^{(b-c)} (b+c-a) (b-c) \cdot x^{(c-a)} (c+a-b) \\
 & \quad (c+a-b) (c-a) \cdot x^{(a-b)} (a+b-c) (a-b) \\
 & = x^{(b^2 - c^2 + c^2 - a^2 + a^2 - b^2)} \cdot x^{-a} (b-c) - b (c-a) - c (a-b) \\
 & = (x^0 \times x^0) = (1 \times 1) = 1.
 \end{aligned}$$

$$\begin{aligned}
 91. \quad & \text{Given Exp.} = x^{(a-b)} (a+b) \cdot x^{(b-c)} (b+c) \cdot x^{(c-a)} (c+a) \\
 & = x^{(a^2 - b^2)} \cdot x^{(b^2 - c^2)} \cdot x^{(c^2 - a^2)} \\
 & = x^{(a^2 - b^2 + b^2 - c^2 + c^2 - a^2)} = x^0 = 1.
 \end{aligned}$$

$$\begin{aligned}
 92. \quad & \text{Given Exp.} = \{x^{(a-b)}\}^{\frac{1}{ab}} \cdot \{x^{(b-c)}\}^{\frac{1}{bc}} \cdot \{x^{(c-a)}\}^{\frac{1}{ca}} \\
 & = x^{\frac{(a-b)}{ab}} \cdot x^{\frac{(b-c)}{bc}} \cdot x^{\frac{(c-a)}{ca}} \\
 & = x^{\left\{\frac{(a-b)}{ab} + \frac{(b-c)}{bc} + \frac{(c-a)}{ca}\right\}} \\
 & = x^{\left\{\frac{1}{b} - \frac{1}{a} + \frac{1}{c} - \frac{1}{b} + \frac{1}{a} - \frac{1}{c}\right\}} = x^0 = 1.
 \end{aligned}$$

$$\begin{aligned}
 93. \quad & \text{Given Exp.} = \left(\frac{xy+1}{y}\right)^a \cdot \left(\frac{xy-1}{y}\right)^b \\
 & \left(\frac{xy+1}{x}\right)^a \cdot \left(\frac{xy-1}{x}\right)^b \\
 & = \frac{(xy+1)^a \cdot (xy-1)^b \cdot x^a \cdot x^b}{(xy+1)^a \cdot (xy-1)^b \cdot y^a \cdot y^b} \\
 & = \frac{x^{a+b}}{y^{a+b}} = \left(\frac{x}{y}\right)^{a+b}.
 \end{aligned}$$

$$\begin{aligned}
 94. \quad & \text{Given Exp.} = x^{\frac{b+c}{(a-b)(c-a)}} \cdot x^{\frac{c+a}{(a-b)(b-c)}} \cdot x^{\frac{a+b}{(b-c)(c-a)}} \\
 & = x^{\frac{(b+c)(b-c) + (c+a)(c-a) + (a+b)(a-b)}{(a-b)(b-c)(c-a)}} \\
 & = x^{\frac{(b^2 - c^2) + (c^2 - a^2) + (a^2 - b^2)}{(a-b)(b-c)(c-a)}} = x^0 = 1.
 \end{aligned}$$

$$\begin{aligned}
 95. \quad & \text{Let } x^p = y^q = z^r = k. \text{ Then, } x = k^{\frac{1}{p}}, y = k^{\frac{1}{q}}, z = k^{\frac{1}{r}}. \\
 & \therefore xyz = 1 \Rightarrow k^{\frac{1}{p}} \cdot k^{\frac{1}{q}} \cdot k^{\frac{1}{r}} = 1 = k^0 \\
 & \Rightarrow k^{\left(\frac{1}{p} + \frac{1}{q} + \frac{1}{r}\right)} = k^0 \Rightarrow \frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 0.
 \end{aligned}$$

96. Let $a^x = b^y = c^z = k$.

Then, $a = k^{\frac{1}{x}}, b = k^{\frac{1}{y}}$ and $c = k^{\frac{1}{z}}$.

$$\therefore b^2 = ac \Leftrightarrow \left(k^{\frac{1}{y}}\right)^2 = k^{\frac{1}{x}} \times k^{\frac{1}{z}} \Leftrightarrow k^{\left(\frac{2}{y}\right)} = k^{\left(\frac{1}{x} + \frac{1}{z}\right)}$$

$$\therefore \frac{2}{y} = \frac{(x+z)}{xz} \Leftrightarrow \frac{y}{2} = \frac{xz}{(x+z)} \Leftrightarrow y = \frac{2xz}{(x+z)}.$$

97. $a^1 = c^z = (b^y)^z = b^{yz} = (a^x)^{yz} = a^{xyz} \Rightarrow xyz = 1$.

98. $2^x = 4^y = 8^z \Leftrightarrow 2^x = 2^{2y} = 2^{3z} \Leftrightarrow x = 2y = 3z$.

$$\begin{aligned} \therefore \frac{1}{2x} + \frac{1}{4y} + \frac{1}{6z} &= \frac{24}{7} \Leftrightarrow \frac{1}{6z} + \frac{1}{6z} + \frac{1}{6z} = \frac{24}{7} \\ &\Leftrightarrow \frac{3}{6z} = \frac{24}{7} \Leftrightarrow z = \left(\frac{3}{6} \times \frac{7}{24}\right) = \frac{7}{48}. \end{aligned}$$

99. $8 = 7^d = (6^c)^d = 6^{cd} = (5^b)^{cd} = 5^{bcd} = (4^a)^{bcd} = 4^{abcd}$

$$\Rightarrow 4^{abcd} = 8 \Rightarrow (2^2)^{abcd} = 2^3 \Rightarrow 2abcd = 3$$

$$\Rightarrow abcd = \frac{3}{2}.$$

$$\begin{aligned} 100. \text{ Given Exp. } &= \frac{1}{1+a+b^{-1}} + \frac{1}{1+b+c^{-1}} + \frac{1}{1+c+a^{-1}} \\ &= \frac{1}{1+a+b^{-1}} + \frac{b^{-1}}{b^{-1}+1+b^{-1}c^{-1}} + \frac{a}{a+ac+1} \\ &= \frac{1}{1+a+b^{-1}} + \frac{b^{-1}}{1+b^{-1}+a} + \frac{a}{a+b^{-1}+1} \\ &= \frac{1+a+b^{-1}}{1+a+b^{-1}} = 1. \end{aligned}$$

$$[\because abc = 1 \Rightarrow (bc)^{-1} = a \Rightarrow b^{-1}c^{-1} = a \text{ and } ac = b^{-1}]$$

$$\begin{aligned} 101. \sqrt{a^{-1}b} \cdot \sqrt{b^{-1}c} \cdot \sqrt{c^{-1}a} &= (a^{-1})^{\frac{1}{2}} \cdot b^{\frac{1}{2}} \cdot (b^{-1})^{\frac{1}{2}} \cdot c^{\frac{1}{2}} \cdot (c^{-1})^{\frac{1}{2}} \cdot a^{\frac{1}{2}} \\ &= (a^{-1}a)^{\frac{1}{2}} \cdot (b \cdot b^{-1})^{\frac{1}{2}} \cdot (c \cdot c^{-1})^{\frac{1}{2}} \\ &= (1)^{\frac{1}{2}} \cdot (1)^{\frac{1}{2}} \cdot (1)^{\frac{1}{2}} = (1 \times 1 \times 1) = 1. \end{aligned}$$

$$102. 3^x - y = 27 = 3^3 \Leftrightarrow x - y = 3 \quad \dots (i)$$

$$3^x + y = 243 = 3^5$$

$$\Leftrightarrow x + y = 5 \quad \dots (ii)$$

On solving (i) and (ii), we get $x = 4$.

103. Let $x^y = y^x = k$.

Then, $x = k^{\frac{1}{y}}$ and $y = k^{\frac{1}{x}}$.

$$\begin{aligned} \therefore \left(\frac{x}{y}\right)^{\frac{x}{y}} &= \left(\frac{k^{\frac{1}{y}}}{k^{\frac{1}{x}}}\right)^{\frac{x}{y}} = k^{\left[\left(\frac{1}{y} - \frac{1}{x}\right) \frac{x}{y}\right]} = k^{\left(\frac{x-y}{xy}\right) \frac{x}{y}} = k^{\left(\frac{x-y}{y^2}\right)} \\ &= (x^y)^{\left(\frac{x-y}{y^2}\right)} = x^{\left(\frac{x-y}{y}\right)} = x^{\left(\frac{x}{y} - 1\right)}. \end{aligned}$$

$$104. 4^x + y = 1 = 4^0 \Leftrightarrow x + y = 0 \quad \dots (i)$$

$$4^x - y = 4 = 4^1 \Rightarrow x - y = 1 \quad \dots (ii)$$

Adding (i) and (ii), we get : $2x = 1$ or $x = \frac{1}{2}$.

Putting $x = \frac{1}{2}$ in (i), we get : $y = -\frac{1}{2}$.

$$105. 2^{2x-1} + 4^x = 2^{x-\frac{1}{2}} + 2^{x+\frac{1}{2}} \Leftrightarrow 2^{2x-1} + 2^{2x} = 2^{x-\frac{1}{2}} + 2^{x+\frac{1}{2}}$$

$$\Leftrightarrow 2^{(2x-1)} (1+2) = 2^{\left(x-\frac{1}{2}\right)} (1+2)$$

$$\Leftrightarrow 2^{(2x-1)} = 2^{\left(x-\frac{1}{2}\right)} \Leftrightarrow 2x-1 = x-\frac{1}{2} \Leftrightarrow x = \frac{1}{2}.$$

$$106. 3^{2x-y} = 3^x + y = \sqrt{3^3} = 3^{\frac{3}{2}} \Leftrightarrow 2x-y = \frac{3}{2} \text{ and } x+y = \frac{3}{2}$$

$$\Leftrightarrow 3x = \frac{3}{2} + \frac{3}{2} = 3 \Leftrightarrow x = 1.$$

$$\therefore y = \left(\frac{3}{2} - 1\right) = \frac{1}{2}.$$

$$107. \text{ Let } 3^x = 5^y = 45^z = k. \text{ Then, } 3 = k^{\frac{1}{x}}, 5 = k^{\frac{1}{y}}, 45 = k^{\frac{1}{z}}.$$

$$45 = 3^2 \times 5$$

$$\Leftrightarrow \frac{1}{k^z} = \left(k^{\frac{1}{x}}\right)^2 \cdot \left(k^{\frac{1}{y}}\right) = k^{\frac{2}{x}} \cdot k^{\frac{1}{y}} = k^{\left(\frac{2}{x} + \frac{1}{y}\right)}$$

$$\Leftrightarrow \frac{1}{z} = \frac{2}{x} + \frac{1}{y} \Leftrightarrow \frac{2}{x} = \frac{1}{z} - \frac{1}{y}.$$

$$108. 2^x = 8^{y+1} \Leftrightarrow 2^x = (2^3)^{y+1} = 2^{(3y+3)} \quad \dots (i)$$

$$\Leftrightarrow x = 3y + 3 \Leftrightarrow x - 3y = 3$$

$$9^y = 3^{x-9} \Leftrightarrow (3^2)^y = 3^{x-9}$$

$$\Leftrightarrow 2y = x - 9 \Leftrightarrow x - 2y = 9 \quad \dots (ii)$$

Subtracting (i) from (ii), we get: $y = 6$. Putting $y = 6$ (i), we get $x = 21$.

$$\therefore x + y = 21 + 6 = 27.$$

$$109. 2^{0.7x} \cdot 3^{-1.25y} = \frac{8\sqrt{6}}{27}$$

$$\Leftrightarrow \frac{2^{0.7x}}{3^{1.25y}} = \frac{2^3 \cdot 2^{\frac{1}{2}} \cdot 3^{\frac{1}{2}}}{3^3} = \frac{2^{\left(3+\frac{1}{2}\right)}}{3^{\left(3-\frac{1}{2}\right)}} = \frac{2^{\frac{7}{2}}}{3^{\frac{5}{2}}} = \frac{2^{3.5}}{3^{2.5}}.$$

$$\therefore 0.7x = 3.5 \Rightarrow x = \frac{3.5}{0.7} = 5 \text{ and } 1.25y = 2.5$$

$$\Rightarrow y = \frac{2.5}{1.25} = 2.$$

$$110. r = (2a)^{2b} = 2^{2b} \times a^{2b} = (2^2)^b \times (a^b)^2 = 4^b \times (a^b)^2.$$

Also, $r = a^b \times x^b$.

$$\therefore a^b \times x^b = 4^b \times (a^b)^2 \Leftrightarrow x^b = 4^b \times a^b = (4a)^b \Leftrightarrow x = 4a.$$

111. L.C.M. of 2, 3, 4, 6 is 12.

$$\sqrt{2} = 2^{\frac{1}{2}} = 2^{\left(\frac{1}{2} \times \frac{6}{6}\right)} = 2^{\frac{6}{12}} = (2^6)^{\frac{1}{12}} = (64)^{\frac{1}{12}} = \sqrt[12]{64}.$$

$$\sqrt[3]{3} = 3^{\frac{1}{3}} = 3^{\left(\frac{1}{3} \times \frac{4}{4}\right)} = 3^{\frac{4}{12}} = (3^4)^{\frac{1}{12}} = (81)^{\frac{1}{12}} = \sqrt[12]{81}.$$

$$\sqrt[4]{4} = 4^{\frac{1}{4}} = 4^{\left(\frac{1}{4} \times \frac{3}{3}\right)} = 4^{\frac{3}{12}} = (4^3)^{\frac{1}{12}} = (64)^{\frac{1}{12}} = \sqrt[12]{64}.$$

$$\sqrt[6]{6} = 6^{\frac{1}{6}} = 6^{\left(\frac{1}{6} \times \frac{2}{2}\right)} = 6^{\frac{2}{12}} = (6^2)^{\frac{1}{12}} = (36)^{\frac{1}{12}} = \sqrt[12]{36}.$$

Clearly, $\sqrt[12]{81}$ i.e., $\sqrt[3]{3}$ is the greatest.

112. L.C.M of 2, 3, 4, 6 is 12.

$$\sqrt{2} = 2^{\frac{1}{2}} = 2^{\left(\frac{1}{2} \times \frac{6}{6}\right)} = 2^{\frac{6}{12}} = (2^6)^{\frac{1}{12}} = (64)^{\frac{1}{12}} = \sqrt[12]{64}.$$

$$\sqrt[3]{3} = 3^{\frac{1}{3}} = 3^{\left(\frac{1}{3} \times \frac{2}{2}\right)} = 3^{\frac{2}{6}} = (3^2)^{\frac{1}{6}} = (9)^{\frac{1}{6}} = \sqrt[6]{9}.$$

$$\sqrt[3]{4} = 4^{\frac{1}{3}} = 4^{\left(\frac{1}{3} \times \frac{4}{4}\right)} = 4^{\frac{4}{12}} = (4^4)^{\frac{1}{12}} = (256)^{\frac{1}{12}} = \sqrt[12]{256}.$$

$$\sqrt[4]{5} = 5^{\frac{1}{4}} = 5^{\left(\frac{1}{4} \times \frac{3}{3}\right)} = 5^{\frac{3}{12}} = (5^3)^{\frac{1}{12}} = (125)^{\frac{1}{12}} = \sqrt[12]{125}.$$

Clearly, $\sqrt[12]{256}$ i.e., $\sqrt[3]{4}$ is the greatest.

113.

114. $x = 5 + 2\sqrt{6} = 3 + 2 + 2\sqrt{6} = (\sqrt{3})^2 + (\sqrt{2})^2 + 2 \times \sqrt{3} \times \sqrt{2}$
 $= (\sqrt{3} + \sqrt{2})^2.$

Also, $(x - 1) = 4 + 2\sqrt{6} = 2(2 + \sqrt{6}) = 2\sqrt{2}(\sqrt{2} + \sqrt{3}).$

$$\therefore \frac{(x-1)}{\sqrt{x}} = \frac{2\sqrt{2}(\sqrt{3} + \sqrt{2})}{(\sqrt{3} + \sqrt{2})} = 2\sqrt{2}.$$

115. $x^{\frac{1}{3}} + y^{\frac{1}{3}} = z^{\frac{1}{3}} \Rightarrow \left(x^{\frac{1}{3}} + y^{\frac{1}{3}}\right)^3 = \left(z^{\frac{1}{3}}\right)^3$

$$\Rightarrow x + y + 3x^{\frac{1}{3}}y^{\frac{1}{3}}\left(x^{\frac{1}{3}} + y^{\frac{1}{3}}\right) = z$$

$$\Rightarrow x + y + 3x^{\frac{1}{3}}y^{\frac{1}{3}}z^{\frac{1}{3}} = z$$

$$\Rightarrow x + y - z = -3x^{\frac{1}{3}}y^{\frac{1}{3}}z^{\frac{1}{3}}$$

$$\Rightarrow (x + y - z)^3 = \left(-3x^{\frac{1}{3}}y^{\frac{1}{3}}z^{\frac{1}{3}}\right)^3$$

$$\Rightarrow (x + y - z)^3 = -27xyz \Rightarrow (x + y - z)^3 + 27xyz = 0.$$

116. $x = 2 + 2^{\frac{2}{3}} + 2^{\frac{1}{3}} \Rightarrow (x - 2) = 2^{\frac{2}{3}} + 2^{\frac{1}{3}}$

$$\Rightarrow (x - 2)^3 = \left(2^{\frac{2}{3}} + 2^{\frac{1}{3}}\right)^3$$

$$= 2^2 + 2 + 3 \cdot 2^{\frac{2}{3}} \cdot 2^{\frac{1}{3}} \left(2^{\frac{2}{3}} + 2^{\frac{1}{3}}\right)$$

$$\Rightarrow (x - 2)^3 = 6 + 6(x - 2) = 6 + 6x - 12$$

$$\Rightarrow (x - 2)^3 = 6x - 6 \Rightarrow x^3 - 8 - 6x(x - 2) = 6x - 6$$

$$\Rightarrow x^3 - 8 - 6x^2 + 12x = 6x - 6 \Rightarrow x^3 - 6x^2 + 6x = 2.$$

117. Given expression $(-2)^5 \times (2)^{-5} \times (3)^3$

$$\frac{(-2)^5}{2^5} \times (3)^3 \quad \left\{ \because a^{-m} = \frac{1}{a^m} \right\}$$

$$\frac{(-1)^5 (2)^5 \times (3)^3}{(2^5)} = (-3)^3 = -27$$

118. Expression = $\frac{(10)^{100}}{(5)^{75}}$

$$= \frac{(2 \times 5)^{100}}{(5)^{75}} = \frac{(2)^{100} \times (5)^{100}}{(5)^{75}} = 2^{100} \times \frac{5^{100}}{5^{75}} = 2^{100} \times 5^{(100-75)}$$

$$\left\{ \because \frac{a^m}{a^n} = a^{m-n} \right\}$$

$$= 2^{100} \times 5^{25}$$

$$= 2^{25} \times 5^{25} \times 2^{75} \quad \left\{ \because a^m \times a^n = a^{m+n} \right\}$$

$$= (10)^{25} \times 2^{75} \quad \left\{ \because a^m \times b^m = ab^m \right\}$$

119. Expression = $\sqrt{\sqrt{2} \times \sqrt{3}}$

$$= (\sqrt{2} \times \sqrt{3})^{\frac{1}{2}} = \left(2^{\frac{1}{2}} \times 3^{\frac{1}{2}}\right)^{\frac{1}{2}}$$

$$\left\{ \because a^m \times b^m = ab^m \right\}$$

$$= (6)^{\frac{1}{2} \times \frac{1}{2}} = (6)^{\frac{1}{4}}$$

$$\left\{ \because (a^m)^n = a^{mn} \right\}$$

120. $21^2 \times 21^{6.5} = 21^{12.4}$

$$\Rightarrow 21^{7+6.5} = 21^{12.4}$$

$$\Rightarrow ? + 6.5 = 12.4$$

$$\Rightarrow ? = 12.4 - 6.5 = 5.9$$

121. $\frac{5.4 + 3 \times 16 + 2}{18 + 5 \times 6 + 3}$

$$= \frac{\frac{5.4}{3} \times \frac{16}{2}}{\frac{18}{5} \times \frac{6}{3}} = \frac{1.8 \times 8}{3.6 \times 2} = 2$$

122. $(32 \times 10^{-5})^2 \times 64 \div (2^{16} \times 10^{-4}) = 10^?$

$$\Rightarrow (2^5 \times 10^{-5})^2 \times 2^6 \div (2^{16} \times 10^{-4}) = 10^? \quad \left\{ \because (a^m)^n = a^{mn} \right\}$$

$$\Rightarrow \frac{2^{10} \times 10^{-10} \times 2^6}{2^{16} \times 10^{-4}} = 10^? \quad \left\{ \because a^m \times a^n = a^{m+n} \right\}$$

$$\Rightarrow \frac{2^{16} \times 10^4}{2^{16} \times 10^{10}} = 10^? \quad \left\{ \because a^{-m} = \frac{1}{a^m} \right\}$$

$$\Rightarrow 10^{4-10} = 10^?$$

$$\Rightarrow 10^{-6} = 10^?$$

$$\Rightarrow ? = -6$$